

## Laboratory notes

### Experimental cell biology

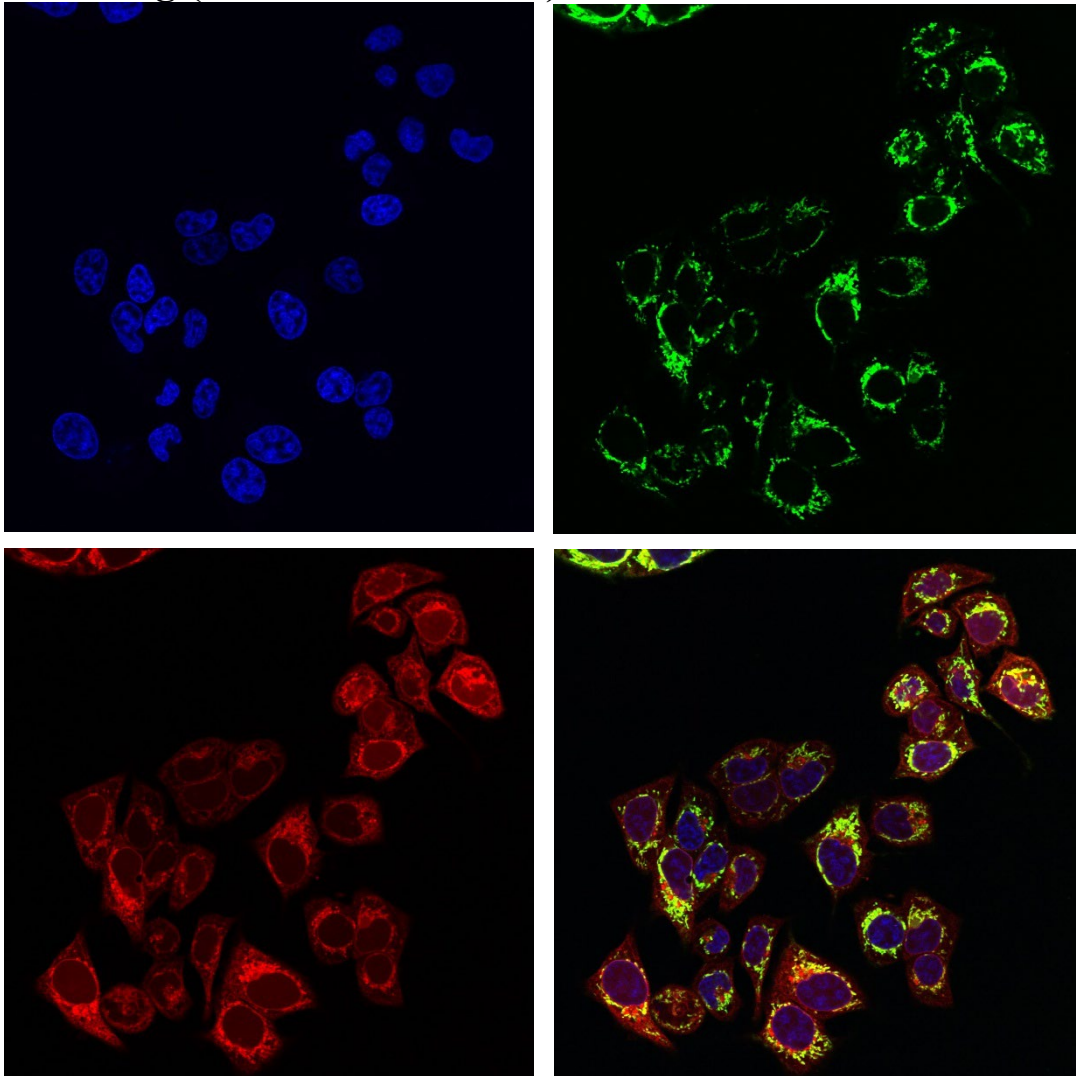
June 2026

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Mitochondria staining using MitoTracker Green (green), actin-skeleton staining using Rhodamine-Phalloidin (red) with the nucleus counterstained using Hoechst 33342 (blue). Courtesy of the Blue team of Experimental Cell Biology 2024.

Relevant books:

Celledyrkning. En praktisk håndbog i dyrkning af mammaie celler. Vivi Kielberg, Nils Brünner, Per Briand. Gads forlag. ISBN 87-12-03747-8

Culture of Animal Cells. A manual of basic technique and specialized applications. R. Ian Freshney. Wiley-Blackwell 6<sup>th</sup> Ed. 2010, ISBN 978-0-470-52812-9.

Basic cell culture. J.M. Davis. Oxford university press. ISBN 0-19-963853-5

Good books, but not required. Borrow from the University library.

Additional, there are a long list of illustrative video on YouTube

## Foreword

The course “Experimental Cell Biology” is now running for a new year, and we hope that you will get good experience with various cell growing techniques, introduced to you at the course. There are many techniques that we cannot show you (because of lacking time), but we hope that these short exercises will make you equipped for your future scientific *in vitro* work.

We are introducing you to grow different mammalian cell lines, show you the methods of counting cells / measuring active metabolic cells, measuring cell cycle distribution, introduce you to the transfection as well as immunocytochemistry. You will try the different techniques twice to increase your confidence in and understanding of the individual techniques. Thus, first week includes only a ‘small setup’ and second week a more ‘complete or complicated’ setup of a typical cell culture experiment.

You will have to submit a pre-report of each experiment after the first week of work. Based on our feedback, you may prepare much better final reports based on all the data when the two weeks in the lab are done.

June 2026

Ole Vang, Course coordinator

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## Introduction to the course

Experimental cell biology work is central to understanding the mechanisms of eukaryotic cell biology. Growing cells, manipulating them and analyzing the effects have identified many central biological processes.

It is our hope that this experimental course will give the students both basic and advanced knowledge in the experimental work that they may bring into their projects and further in their positions after getting their master's degree.

In this course, we will culture the two human colonocyte cell lines:

- Caco-2 (atcc.org: HTB-37 <https://www.lgcstandards.com/DK/en/Caco-2-Colon-Adenocarcinoma-Human-Homo-sapiens-p/ATCC-HTB-37>) and
- DLD-1 (atcc.org: CCL-221 <https://www.lgcstandards.com/DK/en/DLD-1-Colon-Cancer-Human-p/ATCC-CCL-221>)

which will give you the sense of different growth characteristics of difference cell lines.

### Exercises:

**Exercise 1:** General cell culture techniques, sterile technique, time- and dose-response effects, cell counting, cell size measurements (OV)

**Exercise 2:** Cell cycle analysis using flow cytometry and cell number determination in 96-well plates (Hoechst staining) (CMTA)

**Exercise 3:** Transfection and reporter-gene analysis (SBJ)

**Exercise 4:** Fluorescent imaging and staining of cells; hygiene test (OV)

**For each exercise, there is a small setup for week 1 (with a pre-report) and a larger setup for week 2. Each group prepares one final report for each exercise covering the data from both weeks.**

Besides performing exercises, quizzes, and reports, every group has to

- Log everything in their lab notebook: Write about plans, calculations, raw data and calculated results and conclusions. Fill in the notebook according to the instructions on pages 14-15. When you submit your exercise reports (pre-report and final reports) to the teachers then also enclose scans of your lab notebook.
- Include a report front sheet, which is available in the Moodle folder for the course.

- Maintain the Caco-2 and the DLD-1 cell lines growing for the entire course without them becoming confluent. Check the cultures every day, morning and afternoon.
- Freeze down and thaw a cell sample and check for viability.
- Clean the LAF bench both Fridays and on Monday June 30<sup>th</sup> / Tuesday July 1<sup>st</sup>.
- Prepare a Power Point presentation of the main results of each exercise to show the other teams on the last day of the course (June 30<sup>th</sup>).

Before you can start the specific exercise, **you must prepare a flow chart** including every step for this day. Check with the sections “For the report” whether you need to include further samples to solve specific questions. Show this flow chart to the instructor and get it approved before you start in the Lab.

The exercises will be performed in the Cell Culture Rooms/LAF bench in 28A2 (when sterile work is required) or in the FBU lab in 28A1 if sterile work is NOT required). Cell counting on the coulter counter and reading of the 96-well plates are taking place in room 28A.2-21. Images are acquired in the microscope room (28A.2-37) or in the confocal microscope room (28AB.0-07). The flow cytometer is situated in in the FBU lab in 28B.1.

There are several practical rules that you must follow:

1. Confer with the note **General use of cell lab** for the use of the cell laboratories in 28A.2, take special notice of the use of the microscope, centrifuge and disposal of waste (solid as well as fluent).
2. Confer with the note **Cleaning of cell lab** concerning cleaning the LAF benches and incubators.
3. The instructors may each only be present in one place at a time. If in doubt, search and find the instructor to get answers to your questions.
4. Make yourself aware of the correct use of gloves.

Further:

1. You must bring and wear a **clean lab coat** and **indoor shoes** to reduce the risk of contamination. It is not necessary to buy new **shoes**: Launder an old pair of sneakers and bring these.

## Schedule

Every morning at 8:30:

- Meeting place: Meeting room 28A.2-11
- One member of each team goes to cell lab to take out media, PBS/Sodium Citrate and trypsin from fridge (28A.2).
- Overview of each day's exercises is given by OV or SBJ
- Preparation of flow charts of the day – must be approved by OV or SBJ
- Arrange time point for cake and coffee break (sometimes with quiz)

| Day                                     | Date | Exp 1<br>Cell culture<br>Dose-response                                                                                                  | Exp 2<br>Cell cycle and<br>Viability                                        | Exp 3<br>Transfection<br>Reporter-gene        | Exp 4<br>Microscopy<br>Staining /<br>Hygiene                                                                    |
|-----------------------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Friday<br>We start at<br>13:00          | 12   | Introduction to course<br>Practice trypsinization and cell counting (microscope)<br>Do calculation for seeding (Exp 1+2+3+4) for Monday |                                                                             |                                               |                                                                                                                 |
| Mon<br>We start<br>every day at<br>8:30 | 15   | Trypsinize, seed<br>for passage +<br>seed for Exp 1, 2,<br>3 and 4<br>Lecture by <b>OV</b> on<br>cell culture                           | Seed for cell cycle and viability assays                                    | Seed cells for transfection                   | Seed cells in Ibidi Dishes                                                                                      |
| Tues                                    | 16   | Treat with Resv & NaBu                                                                                                                  | Treat cells with NaBu                                                       | Transfect cells with plasmids                 | Treat cells with NaBu/ Resv                                                                                     |
| Wed                                     | 17   | Perhaps passage<br>Freeze down<br>DLD-1<br>Lecture by <b>OV</b> on<br>data analysis                                                     | Cell cycle assay with FACS<br>Lecture by <b>OV</b> on cell viability assays | Change media                                  | Coat cover slips<br><br>Hygiene test                                                                            |
| Thur<br><b>LONG DAY!</b>                | 18   | Image and count cells<br>Perhaps passage<br>Seed for Exp 4                                                                              | Perform viability assay.                                                    | Measure luciferase activity                   | Seed cells on cover slips (for week 2)                                                                          |
| Fri<br><b>LONG DAY!</b>                 | 19   | Data analysis<br>Passage cells<br>Clean LAF hood and incubators                                                                         | Data analysis                                                               | Seed cells for luciferase proliferation assay | Treat with Resv and NaBu (for week 2)<br>Stain and image live fluorescence.<br>Acquire images using microscopes |



|                          |    |                                                                                   |                                          |                                                                                                            |                                                                                                |
|--------------------------|----|-----------------------------------------------------------------------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Mon                      | 22 | Hand in a pre-report at 8:30 <b>OV</b>                                            | Hand in a pre-report at 8:30 <b>CMTA</b> | Hand in a pre-report at 8:30 <b>SBJ</b>                                                                    | Hand in a pre-report at 8:30 <b>OV</b>                                                         |
|                          |    | <b>Trypsinize, seed for passage seed for Exp 1, 2 and 3</b>                       | Seed for cell cycle and viability assays | Seed cells for transfection<br><br>Measure luciferase activity                                             | Fix and stain with DAPI and Rhodamine-Phalloidin<br>Lecture by <b>OV</b> on microscope methods |
| Tues                     | 23 | Treat with Resv & NaBu                                                            | Treat cells with Resv & NaBu             | Transfect cells with plasmids<br>Measure luciferase activity<br><br>Lecture by <b>SBJ</b> on data analysis | Acquire images on microscopes                                                                  |
| Wed                      | 24 | Perhaps passage<br>Thaw cells from N <sub>2</sub><br>Lecture by <b>OV</b> on Resv | Cell cycle analysis with FACS            | Change media<br><br>Measure luciferase activity                                                            | Acquire images on microscopes<br>Data analysis                                                 |
| Thur<br><b>LONG DAY!</b> | 25 | Image and count cells                                                             | Perform viability assay.                 | Measure luciferase activity                                                                                |                                                                                                |
| Fri<br><b>LONG DAY!</b>  | 26 | Trypsinize and passage cells<br>Data analysis<br>Prepare report                   | Data analysis<br><br>Prepare report.     | Data analysis<br><br>Prepare report.<br>Hand-in report at noon to <b>SBJ</b>                               | Data analysis<br><br>Prepare report.<br>Hand-in report at 4 pm to <b>OV</b>                    |
| Mon                      | 29 | Dispose.<br>Clean LAF hood and incubators.<br>Hand-in report by 4 pm to <b>OV</b> | Hand-in report at noon to <b>CMTA</b>    | Get report back.                                                                                           | Get report back.                                                                               |
| Tue                      | 30 | Get report back.                                                                  | Get report back                          |                                                                                                            |                                                                                                |
|                          |    | Presentation of results for the other groups.                                     |                                          |                                                                                                            |                                                                                                |
|                          |    | When you get all the reports approved, submit your reports to eksamen.ruc.dk.     |                                          |                                                                                                            |                                                                                                |
|                          |    | Visit at Novo in Kalundborg, back at 17.00 at the latest                          |                                          |                                                                                                            |                                                                                                |

## Introduction to cell work

With all the precautions and strict sterile demands necessary for working with cell cultures, the field can seem a bit overwhelming for the first time student. However, the truth is that if you can change media and resuspend your cells without contaminating your culture, you can quickly learn the basic aspects of working with cells.

### Primary Cultures

Primary cultures are derived directly from excised, normal animal tissue and cultured. The preparation of primary cultures is labor intensive, and they can be maintained *in vitro* only for a limited period, as the differentiated cells normally do not proliferate. During their relatively limited life span, primary cells usually retain many of the differentiated characteristics of the cell *in vivo*.

### Continuous Cultures

Continuous cultures are comprised of a single cell type that can be serially propagated in culture either for a limited number of cell divisions (approximately thirty sub cultivations) or otherwise indefinitely. Cell lines of a finite life are usually diploid and maintain some degree of differentiation. The fact that such cell lines senesce after approximately thirty cycles of sub cultivations means it is essential to establish a system of Master and Working banks to maintain such lines for long periods. Mouse embryonic fibroblasts (MEFs) are examples of such continuous cultures with limited proliferative capacity.

Continuous cell lines that can be propagated indefinitely generally have this ability because they have been transformed into tumor cells. Tumor cell lines are often derived from actual clinical tumors, but transformation may also be induced using viral oncogenes or by chemical treatments. Transformed cell lines present the advantage of almost limitless availability, but the disadvantage of having retained very little of the original *in vivo* characteristics. Most cell lines are transformed.

### Culture Morphology

Morphologically cell cultures take one of two forms, growing either in suspension (as single cells or small free-floating clumps) or as a monolayer that is attached to the tissue culture flask. The form taken by a cell line reflects the tissue from which it was derived e.g. cell lines derived from blood (leukaemia, lymphoma) tend to grow in suspension whereas cells derived from solid tissue (lungs, kidney, colon) tend to grow as monolayers.

### **Maintaining cells in culture**

Cells are grown and maintained at an appropriate temperature and gas mixture (typically, 37°C, 5% CO<sub>2</sub>) in a cell incubator. Culture conditions vary widely for each cell type, and variation of conditions for a particular cell type can result in different phenotypes.

Aside from temperature and gas mixture, the most varied factor in culture systems is the growth medium. Recipes for growth media can vary in pH, glucose concentration, growth factors, and the presence of other nutrient components. The growth factors used to supplement media often derive from animal blood, such as calf serum.

Some cells naturally live without attaching to a surface, such as cells that exist in the bloodstream. Others require a surface, such as most cells derived from solid tissues. Cells grown unattached to the surface are referred to as suspension cultures. Other adherent cultures cells are grown on tissue culture plastic, which may be coated with extracellular matrix components to increase its adhesion properties and provide other signals needed for growth. Loosely adherent cells may have their attachment to plastic or glass surfaces vastly increased by coating the surface (in Exp 4 you will coat glass cover slips with poly-lysine).

As cells grow in culture, we can observe three distinct phases. The first is a Lag Phase, usually no more than 24 hours in length, and during which there is little or no increase in cell number. During this time, the cells are "conditioning" the media, undergoing internal cytoskeletal and enzyme changes and adjusting to the new media. This phase is very different in length for different cell lines (Caco-2 and DLD-1 are fast whereas the insulin secreting cells INS-1E are slow).

In the next phase the cell number increases exponentially (Log phase). This growth will continue as long as there is sufficient nutrient and space to support the increasing cell number. Eventually some critical nutrient will become limiting.

The final phase is the Plateau Phase. During this phase, the number of cells remains constant (although not necessarily viable). At some point, the cells will die unless sub-cultured or fresh media is added.

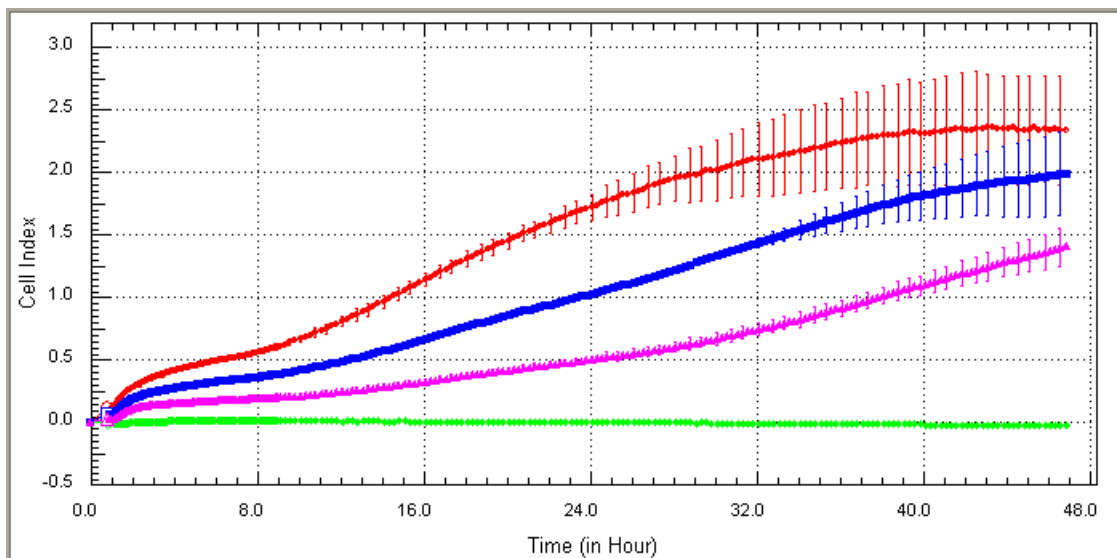


Figure: Xcelligence graph of INS-1E cells growing over 48 hrs. Wells were seeded with 60000 (red), 40000 (blue), 20000 (pink) and 0 cells (green). Note the lag time in cell growth which is approximately 12 hrs (red line) to 32 hrs (pink) for these cells depending on seeding density. After about 40 hrs the wells seeded with 60000 cells are confluent and there is no longer an increasing signal.

### Media changes

The purpose of media changes is to replenish nutrients and avoid the buildup of potentially harmful metabolic by-products and dead cells. In the case of suspension cultures, cells can be separated from the media by centrifugation and re-suspended in fresh media. In the case of adherent cultures, the media can be removed directly by aspiration and be replaced.

### Passaging cells

Passaging or splitting cells involves transferring a smaller number of cells into a new vessel. Cells can be cultured for a longer time if they are split regularly, as it avoids the senescence associated with prolonged high cell density. Suspension cultures are easily passaged with a small amount of culture containing a few cells diluted in a larger volume of fresh media. For adherent cultures, cells first need to be detached; this is commonly done with a mixture of trypsin-EDTA, however other enzyme mixes are now available

for this purpose. A small number of detached cells is used to seed a new culture and the rest may be used to seed plates for experiments. One must be aware that too long time of exposure or to high amount of trypsin will significantly decrease the ability of the cells to reattach. The terms 'passaging', 'splitting', 'sub-cultivating', 'trypsinize and seed' all refer to essentially the same process.

### **Cryopreservation of Cell Lines**

The aim of cryopreservation is to enable stocks of cells to be stored to prevent the need to have all cell lines in culture at all times. It is invaluable when dealing with cells of limited life span. The other main advantages of cryopreservation are:

- Reduced risk of microbial contamination
- Reduced risk of cross contamination with other cell lines
- Reduced risk of genetic drift and morphological changes
- Work conducted using cells at a consistent passage number

The basic principle of successful cryopreservation is a slow freeze and quick thaw. Although the precise requirement may vary with different cell lines as a general guide cells should be cooled at a rate of  $-1^{\circ}\text{C}$  to  $-3^{\circ}\text{C}$  per minute and thawed quickly by heating the sample in  $37^{\circ}\text{C}$  water. If this and the additional points given below are followed, then most cell lines should be cryopreserved successfully.

1. Cultures should be healthy with a viability of  $>90\%$  ( $>70\%$  for Caco-2) and no signs of microbial contamination.
2. Cultures should be in exponential phase of cell growth (not confluent).
3. Use a cryoprotectant such as dimethyl sulfoxide (DMSO) or glycerol to help protect the cells from rupture by the formation of ice crystals. Cryoprotectants are usually used in a 10% concentration in complete media.

### **Laboratory safety**

1. Use gloves sensibly: Wear gloves when handling potentially hazardous material, in the cell culture cabinet. Do not wear gloves when touching handles, door-knobs, instruments or similar – these are considered clean, whereas your gloves are considered contaminated
2. Always wear a clean lab-coat dedicated to cell biology work.
3. Use dedicated footwear in the laboratory.
4. Follow usual GMO Class I rules (no food, no drink, no chewing gum, no jewelry in the lab.)

### **Laboratory notebook**

It is important to keep a proper and tidy laboratory notebook. First of all, you will never be able to remember all the little details when you write up your thesis or the manuscript of your article. The notebook is a source for determining which of your perhaps deviating experiments you should trust, or to trouble-shoot why something has stopped working. Furthermore, the laboratory notebook, when properly used, filled in and signed is a good indication of inventorship, hence the industry relies heavily on laboratory notebooks for this purpose (often in an electronic version). See an example of instructions on page 17. Follow these when filling in your own lab notebook.

### **General use of cell lab**

If you are a new user, you must be introduced to the usage of the cell lab by an experienced user. Contact your supervisor.

When reserving a LAF bench or tagging culture flasks or other items as yours, please identify yourself by your email or initials as identification in the cell lab (include the initials of your supervisor). This allows quick identification of users in case of any problems or questions.

### **Daily use of cell lab:**

Always log your actions in the cell lab in the logbook at the desk when you are finished. Your log should contain your name, date of action, cell type used, and if you encountered any problems with contamination and/or devices

If you are the first user of the day, empty the two waste flasks of the pumps and refill with 25 mL of deconex.

When starting the LAF benches, wait 10 minutes before usage to ensure maximal air ventilation. When turning off the LAF benches, please remember that the big LAF bench should always be on half-power.

Clean up after yourself. Used Pasteur pipettes are to be put in one of the two waste containers under the LAF bench. If the container is full, then take it to the autoclave room.

Remember that all materials that have been in contact with cell material **must** be autoclaved. Place items in an autoclave suited bag/container and make sure it gets autoclaved.

### **Cleaning of cell lab**

Weekly cleaning includes:

- Thorough cleaning of the LAF benches first with 10 % deconex and after that 1% Rodalon and 70% Ethanol
- Cleaning of water pumps. Both the waste flask and suction flask should be cleaned. Rinse well use brush
- Refilling the two incubators with autoclaved milli Q water
- Autoclaving of containers with used Pasteur pipettes if they are full (or nearly)

Monthly cleaning includes

- All of the above
- Cleaning of incubators; first with 10 % deconex and after that 1% Rodalon
- Cleaning of centrifuge; first with 10 % deconex and after that 1% Rodalon



## INSTRUCTIONS FOR KEEPING RESEARCH RECORDS

In addition to providing a complete record of your laboratory work which can be understood and repeated by yourself and others, this notebook has been designed to afford maximum patent right protection. Several practices must be followed to give the notebook value as a legal document in possible patent litigation:

1. Enter all data directly into this book; it is permanently bound with numbered pages so that pages can not be substituted or deleted. Insert a piece of carbon paper between each original and duplicate page in turn such that a copy of all dates, data and signatures are made as work progresses. These copies should be removed from the book and given to your group leader. Do not record data elsewhere for transfer into the book. Write in ink. Never make erasures. Thus, the integrity of the record will not be in question.
2. Record sufficient information. All procedures, reagents, apparatus, sketches, conditions, references, etc., should be entered into the book as the work is done. The purpose and significance of the experiment as well as observations, results, and conclusions should be made clear. What may seem trivial at the time may later prove of critical importance. Your entries should be clear and complete enough for someone else who is "skilled in the art" to read and comprehend what has been accomplished. Avoid sweeping negative statements, e.g.: "This procedure is worthless," which could later limit the scope of your claims.
3. Not only is the conception of an invention important, but so is the diligence shown in making a working model or demonstrating that the idea works—"reducing to practice." These two elements of an invention, conception and reduction to practice, must be corroborated by a witness. The records of the inventor(s) are not enough. Thus, each page of the notebook should be read, witnessed, and dated (daily, if possible) by someone who is competent to understand it, but who does not claim to be a co-inventor. Charts, tables, etc., should be complete, and lines should be drawn through any blank spaces prior to witnessing. It may be wise to perform key experiments in front of one or more witnesses. Spectra, charts, etc., should be signed, dated, witnessed, and if they can not be permanently attached to the notebook, they should be referred to with an entry in the book and kept on file. Dates and witnesses can establish your priority of invention.
4. To delete an entry, draw a line through it so that it is still legible. Corrections should be made adjacent to the deleted entry, and they should be initialled and dated by you and the corroborating witness. Changes made after the page has been witnessed should also be initialled and dated by you and the witness. Always use the current date.
5. The notebook and its contents are to be considered confidential and of great value. Exercise every care in preserving them. Report the loss or theft of a research notebook to your group leader immediately.
6. Index the contents and return each book as completed (or when not in use) for filing.
7. New ideas must be recorded and witnessed as they occur to establish priority of invention. Even ideas which do not pertain to the project at hand should be documented in the book.

Keep your research records as if each project were to be patented. Even though the work contained in the book may not result in a patent application, observance of these practices will provide a clear record for reports, publication, or future reference.

Instructions Read and Understood by

Dated

Hansen (signature)  
15/10-07

## Exercise 1 – General cell culture techniques

Confer with the section **Introduction to cell work**.

In this exercise, you will learn how to work aseptically, maintain a cell culture, count cells and acquire phase contrast images. Moreover, you will become familiar with the routine calculations performed to do dose-response titrations, measure cell diameter and area.

In this exercise, you will also learn how to estimate the dose-response effect of a cytostatic compound (resveratrol) and a differentiation-inducing compound (Sodium Butyrate (NaBu) on cell number and cell diameter. You will become familiar with the Coulter counter and phase-contrast microscopy.

### Introduction

#### Aim

Each group receives two T-75 flasks containing Caco-2 AND one T-75 flask with DLD-1 cells and from these you should:

- Subcultivate and maintain growing for the rest of the course. Subcultivate both Caco-2 and DLD-1 cells Mondays and Thursdays.
  - DLD-1 cells grow fast. Hence, if cells show signs of becoming confluent overnight, then they should be splitted before leaving the lab.
  - Split Caco-2 cells before they reach 70% confluency. If Caco-2 cells become more than 80% confluent they start a differentiation process, which will stop cell division, and they will be less efficiently transfected. This effect cannot be reverted, so Caco-2 cells from flasks 80% or more confluent should not be carried forward.
- Freeze down DLD-1 cells and thaw a cell sample and culture again.

## Preparation of media

Note: The portioning described below is done for you in advance. You will receive media (with FBS<sup>1</sup>), PBS and portions of trypsin-EDTA and sodium citrate but will need to aliquot<sup>2</sup> the media yourself. This is solely be done to reduce the risk of contamination.

**Portioning of media:** Trypsin-EDTA 0.05% (Invitrogen, GIBCO), FBS and penicillin/streptomycin 10.000/5000 u/mL (pen/strep) stock solutions are stored at -20 °C. When necessary, a stock solution is defrosted and split into portions:

- Trypsin-EDTA is transferred to 15 mL tubes with 10 mL in each. The portions are kept at -20°C until usage but are stored at 4°C after first use. Trypsin has a limited lifetime. Assume it expires 2 weeks after thawing.
- FBS portions are kept in 50 mL tubes, and exactly 55 mL is used for a 500 mL media solution (10%) in each. If portions are scheduled for use within 6 months they can be stored at 4°C else they are frozen until use.
- Pen/strep solution is transferred to 15 mL tubes with 11 mL in each. The portions are stored at 4 °C.
- Sodium citrate solution can be stored at room temperature.

## Trypsinizing cells

### Materials:

- Trypsin-EDTA 0.05% (ThermoFisher, Invitrogen, GIBCO)
- PBS w/o Ca and Mg (PBSw/o) (PBS: Phosphate buffered saline, major laboratory suppliers, MLS)
- Sodium citrate, sterile (85 mM)
- Supplemented complete growth media (see Table 1.1)

---

<sup>1</sup> FBS: Fetal bovine serum, is the same as fetal calf serum, FCS. <sup>2</sup> Aliquot: Make smaller portions (‘ud-portionere’)

| <b>Table 1.1 Common cell lines and relevant media</b> |                                                     |                                                                          |
|-------------------------------------------------------|-----------------------------------------------------|--------------------------------------------------------------------------|
| <b>Cell line</b>                                      | <b>Growth media</b>                                 | <b>Comment</b>                                                           |
| <b>DLD-1</b>                                          | <b>McCoy 5A + 10% FBS + 1% P/S</b>                  | <b>For T75, seed 0.5-1 million cells if passaged mon+thurs</b>           |
| <b>Caco-2</b>                                         | <b>DMEM + 10% FBS + 1% P/S</b>                      | <b>For T75, seed 0.2-0.5 million cells if passaged mon+thurs</b>         |
| INS-1E                                                | RPMI 1640 + 10% FBS + 1% P/S + 50µM mercaptoethanol | For T75, seed 3 million cells if passaged mon, feed fresh medium wed+fri |
| HaCAT, C2C12, HeLa, HEK293                            | DMEM + 10% FBS + 1% P/S                             | For T75, seed 1-1.5 million cells if passaged mon+thurs                  |
| CRL-1740                                              | RPMI 1640 + 10% FBS + 1% P/S                        |                                                                          |
| DU-145                                                | RPMI 1640 + 10% FBS + 1% P/S                        |                                                                          |
| HCT-116                                               | McCoy 5A + 10% FBS + 1% P/S                         |                                                                          |

**Procedure:**

1. Take a look on <https://youtu.be/CMRKKI9XSDU>
2. Old culture media is removed by suction through a Pasteur pipette. Be careful not to remove adherent cells during the process.
3. Both DLD-1 and Caco-2 cells require 2 washes prior to trypsin treatment: DLD-1 with PBSw/o Ca or Mg and Caco-2 with 85 mM sodium citrate. A suitable amount of luke-warm PBSw/o **OR** 85 mM sodium citrate is applied, and the cells are carefully rinsed (10mL is suitable for a T75 flask).
4. PBSw/o /sodium citrate is removed by suction without touching the cell layer.
  - A suitable amount of fresh trypsin-EDTA is applied and is dispersed over the cells. (1mL usually for a T75 flask, see table below).
5. The culture flask/dish with trypsin-EDTA is put in the incubator for 3 minutes after which the cells are inspected whether they are loosened by slapping the culture flask/dish. If the cells have not been successfully detached, the flask/dish is incubated for additional 3 minutes. The cells must, however, not be exposed to trypsin for more than 8-10 minutes total (for subcultivation).
6. Detached cells are resuspended in media **OR** PBSw/o (unless otherwise stated use about the same volume as for the washing step). End volume: 10mL for a T75 flask

(Table 1.2). Note: Normally complete supplemented media would be used. In some cases when cells are not used for further growing and if we want to reduce cell aggregation, we resuspend in PBSw/o.

7. If the cells are to be used in further experiments, the cells are transferred to a 15 mL centrifuge tube.
8. If further growth is wanted, the cells are split (often 1/5, 1/10 or 1/20 or with a specified number of cells per flask (Table 1.1)) and a suitable volume of fresh culture media is added to the new culture flask/well. If cells are a bit sensitive towards being seeded too thin or too thick, then count them in a hemocytometer/Coulter counter and seed the appropriate amount. E.g. Caco-2 should not be seeded with less than  $2-3 \times 10^5$  cells per T75 flask.

| <b>Table 1.2 Areas and suggested volumens</b> |                                            |                                           |                                      |
|-----------------------------------------------|--------------------------------------------|-------------------------------------------|--------------------------------------|
| <b>Growing area (cm<sup>2</sup>)</b>          | <b>Recommended volume of culture media</b> | <b>Recommended volume of PBS for wash</b> | <b>Recommended volume of trypsin</b> |
| <b>96 well plate (0.32)</b>                   | 100-200 µL/well                            | 200 µL/well                               | -                                    |
| <b>48 well plate (0.95)</b>                   | 300-400 µL/well                            | 300 µL/well                               | -                                    |
| <b>24 well plate (1.9)</b>                    | 0.5-1 mL/well                              | 1 mL/well                                 | 0.2 mL/well                          |
| <b>12 well plate (3.8)</b>                    | 1.5-2 mL/well                              | 1.5 mL/well                               | -                                    |
| <b>6 well plate (9.5)</b>                     | 3.0 mL/well                                | 2.0 mL/well                               | 0.5 mL/well                          |
| <b>3 cm Ø dish (10)</b>                       | 2-3 mL                                     | 2.0 mL                                    | 0.5 mL                               |
| <b>10 cm Ø dish (55)</b>                      | 10 mL                                      | 7.5 mL                                    | 0.75 mL                              |
| <b>T25 (25)</b>                               | 5 mL                                       | 2.0 mL                                    | 0.5 mL                               |
| <b>T75 (75)</b>                               | <b>13 mL</b>                               | <b>10 mL</b>                              | <b>1 mL</b>                          |
| <b>T150 (150)</b>                             | 35 mL                                      | 15 mL                                     | 2.0 mL                               |

Cells may be counted in different ways. Below are two techniques described, Coulter counting and hemocytometer.

## Counting cells (Coulter counter)

### Materials:

- PBSw/o, sterile Sodium citrate, or media,

- 0.9% saline (physiological saline)

**Procedure:**

1. Cells are trypsinized as described above (**Trypsinizing cells**, p19)
2. 250-500  $\mu\text{L}$  of each re-suspended cell culture are transferred to a 1.5 mL Eppendorf tube. NOTE: If you need to further passage your cells then be very careful to work aseptically. Otherwise, this may be done outside the LAF bench.
3. Setup the Coulter counter (MultiSizer 4e, 28A.2-21):
  - Select and load a relevant SOP (Standard Operation Procedure) file
  - Count cells above 9  $\mu\text{m}$  (up to 40  $\mu\text{m}$ )
  - Count 0.5 mL per sample.
  - Gain: 4
  - Make sure that the output is **cell count**.
4. Clean the Coulter counter before usage (Flush function) with 0.9 % saline.
5. Invert your Eppendorf tube with cells twice and take out 100-200  $\mu\text{L}$  and re-suspend in 10 mL 0.9% saline using the plastic cups in laboratory 28A.2-21. The sample is counted in the Coulter counter and the result is noted (See appendix).  
Repeat the counting.
  - A counting file is saved for each counting. The results of these files may be used for further analysis.
6. After the final count, clean the Coulter counter (Flush function) and collect all cell material so it may be destructed (Read notice on Coulter counter).
7. Indicate your use in the logbook (spreadsheet) at the Coulter counter computer.
8. The cell counts are filled in the table and recorded in the Lab NoteBook.
9. The files are saved automatically by MultiSizer 4e software.

|                                         |
|-----------------------------------------|
| <h2>Counting cells (hemocytometer)</h2> |
|-----------------------------------------|

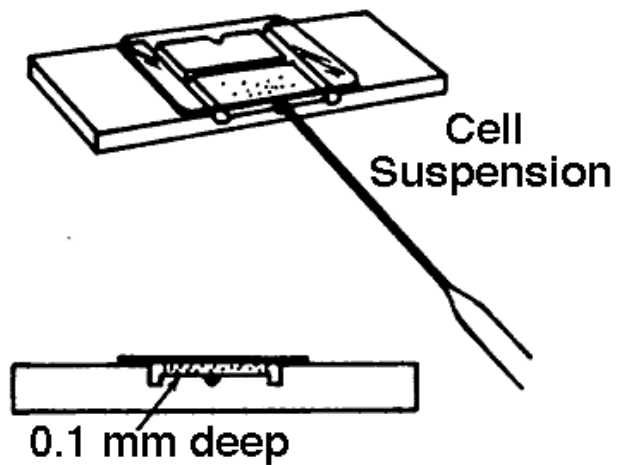
**Materials:**

- PBSw/o or media
- 0.4 % (w/v) trypan blue in PBS (optional)

- Click-counter
- Clean Hemocytometer

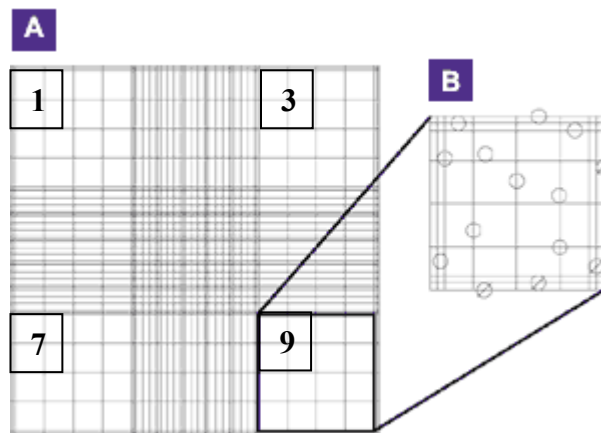
**Procedure:**

1. Cells are trypsinized as described above (**Trypsinizing cells**, p19)
2. Find a hemocytometer (marked “Türk” or “Bürker-Türk”), make sure it is clean and dry.
3. Re-suspend the cell solution and add a droplet (10  $\mu\text{L}$  sample) as shown on the figure to the right.
4. Inspect in the microscope (use 10x objective)
5. Count the number of cells according to figure:



1. Here, **A** indicates one chamber of the hemo-cytometer, containing nine squares (1-9). Each square is 1  $\text{mm}^2$  = 100nl.

2. **B** is an enlarged view of the ninth square, with a sample of cells. In this sample, only count the cells marked as. The cells with diagonal lines through them should not be counted.



3. Count the number of cells in square 1, 3, 7 and 9 in one chamber. Count about 100 cells /  $\text{mm}^2$ .
  - If there are very few cells (< 100 cells/ $\text{mm}^2$ ) count one or more additional squares
  - If there are too many cells make a new dilution with a lower cell density.
4. Count cells in the second counting chamber.
5. Cell density / mL = Cell count (mean of 4 squares) \* 10 000 \* dilution factor

6. Example: Mean count of 4 squares: 55. Suspension was diluted 2-fold before application to the hemocytometer. Concentration of cells:  $55 \text{ cells} \times 10\,000 \text{ mL}^{-1} \times 2 \text{ fold} = 1,1 \times 10^6 \text{ cells/mL}$ . Total amount of cells from a culture flask (often 10 mL): Volume after harvest \* concentration of cells =  $10 \text{ mL} \times 1,1 \times 10^6 \text{ cells/mL} = 1,1 \times 10^7 \text{ cells}$ .

**Note:** Cell viability may be determined by using trypan blue exclusion assay. You do not have to do this every time, but you should try it out during the course.

1. A 100  $\mu\text{L}$  sample of the trypsinized cell suspension is transferred into a 1.5 mL centrifuge tube
2. Mix with 100  $\mu\text{L}$  sample of 0.4 % (w/v) trypan blue
3. Add a droplet of the solution to the hemocytometer and count as described above
4. % viability =  $100 \times (\text{viable cells (White)}) / (\text{viable cells (White)} + \text{non-viable cells (Blue)})$

**Example calculation of cell suspension volume needed:**

You will often need to calculate the volume of cells for seeding, given the specific density of cells determined after harvest.

This is conveniently done using the formula:  $C_1 \times V_1 = C_2 \times V_2$  in some re-arranged version.

Example 1: For a T150 you need to seed  $6 \times 10^6$  cells in a total volume of 30 mL. You have harvested 10 mL of cells with a density of  $2.1 \times 10^6 \text{ cells/mL}$ .

First, find out the volume containing the needed number of cells:

$$\frac{\text{number of cells needed}}{\text{density of cells available}} = \frac{6 \times 10^6 (\text{cells})}{2.1 \times 10^6 (\frac{\text{cells}}{\text{mL}})} = 2,85 \text{ mL}.$$

Thus, 2,85mL of the cell suspension is transferred to the new T150 flask. The amount of new media to add:  $30\text{mL} - 2.85\text{mL} = 27 \text{ mL media}$ .

Example 2: You need, for a number of different setups,  $8 \times 10^6$  cells (8 million cells) **and** need cells to seed for the next passage (1 million cells). You have harvested 10 mL



of cells with a density of  $2.1 \times 10^6$  cells/ mL. Total amount of cells: Volume of suspension \* density:  $10 \text{ mL} \times 2.1 \times 10^6 \text{ cells/ mL} = 21 \times 10^6$  cells – thus, you have enough cells for both experimental setups and further passaging.

When calculating amounts of cells needed for a master mix for several wells, it is a good idea to calculate with a surplus of 15% to accommodate errors in pipetting volumes.

## Freezing DLD-1 cells (Long term storage)

### Materials:

- Trypsin-EDTA 0.05% (ThermoFisher, Invitrogen, GIBCO)
- PBSw/o
- Culture media
- DMSO (Sterile glycerol may also be used)
- Cryo tubes (Nunc, ThermoFischer)
- Mr. Freeze container

### Procedure:

1. Take a look on [https://youtu.be/dXMG\\_R8vh18](https://youtu.be/dXMG_R8vh18).
2. Cells to be frozen must be in good growth (not confluent). Old culture media is removed by suction through Pasteur pipette. Be careful not to remove cells during the process
3. A suitable amount of luke-warm PBSw/o is applied, and the cells are carefully washed
4. PBS is removed by suction.
5. Repeat step 3 and 4.
6. A suitable amount of fresh trypsin-EDTA is applied and is dispersed over the cells (according to Table 1.2).
7. The culture flask with trypsin-EDTA is put in the incubator for 3 minutes after which the cells are loosened by slapping the culture flask/well carefully. If the cells

have not been successfully loosened, the flask/well is incubated for additional 3 minutes. The cells must, however, not be exposed to trypsin for more than 8-10 minutes total.

8. The cells are re-suspended in complete media, centrifuged at 145 G (similar to 1000 rpm to the centrifuge laced in the cell lab) for 3-5 min at room temperature. The media is removed by suction using a Pasteur pipette, and cells are re-suspended in suitable freezing media (Depending on the cell line. See table at p. 26). Calculate appr. 1 mL freezing media per cryotube wanted. Keep cells alive until the procedure has been completed. NOTE: Often cells are frozen at very high densities:  $> 2 \times 10^6$  per cryo-vial. This ensures speedy recovery of cells following thawing.
9. Mark cryotubes with name, cell line, number of cells, passage number and date.
10. 1 mL is transferred to as many marked cryotubes as wanted (you freeze down max 2 vials)
11. The cryotubes are placed in a Mr. Freeze container, which is placed at  $-80^{\circ}\text{C}$  for at least 3 hours (or over-night). After this the tubes are journalized and placed in the cryo-container containing liquid  $\text{N}_2$ . Note date of use on lid of the Mr. Freeze container. The isopropanol within the container has to be changed for every 5 freeze cycles
12. Thaw and grow one of the frozen cultures to ensure that the procedure was successful.

## Thawing cells

### Materials:

- Culture media

### Procedure:

1. Take a look on <https://youtu.be/GhJTn74xxC0>.
2. 15 mL culture media is added to a T75 culture flask. The flask is placed in the incubator.

3. The selected cryotube is taken from the cryo-container and kept on ice. Normal procedure would be: The cell culture vial is found in the cryo-bank registry, and it is noted that the culture vial is removed.
4. The cryotube is defrosted by submerging it in 37°C water. The defrosting should be complete within 1 minute. Note the information on the vial, before the cryotube is cleaned on the outside with 70% EtOH. Moved the cryotube into sterile environment. Then the culture is transferred into the preheated T75 flask with culture media.
5. DLD-1 cells will tolerate this treatment. More delicate cell lines require that the cells are resuspended in 10 mL growth medium and the remaining cryoprotectant is removed by centrifugation (5min@145G), removal of all medium and gentle resuspension in fresh growth medium before seeding.
6. The next day the culture is investigated in a microscope. The degree of adherent cells (or confluence) is noted, and growth media is changed.

**Table 1.3**

| Cell line                  | Freezing media                                                 |
|----------------------------|----------------------------------------------------------------|
| <b>DLD-1</b>               | <b>McCoy 5A + 10% FBS + 1% P/S + 10% Glycerol</b>              |
| Caco-2, HaCAT              | DMEM + 10% FBS + 1% P/S + 10% Glycerol                         |
| INS-1E                     | RPMI 1640 + 10% FBS + 1% P/S + 50µM Mercaptoethanol + 10% DMSO |
| CHO-AAS                    | DMEM + 20% FBS + 10% Glycerol                                  |
| CRL-1740                   | RPMI 1640 + 10% FBS + 1% P/S + 10% Glycerol                    |
| DU-145                     | RPMI 1640 + 10% FBS + 1% P/S + 10% Glycerol                    |
| HCT-116                    | McCoy 5A + 10% FBS + 1% P/S + 10% Glycerol                     |
| HeLa                       | DMEM + 10% FBS + 1% P/S + 10% Glycerol                         |
| HeLa Rho <sup>0</sup>      | DMEM + 10% FBS + 1% P/S + 10% Glycerol + 4 µg/mL Uridine       |
| LNCaP                      | McCoy 5A + 10% FBS + 1% P/S + 10% Glycerol                     |
| Lovo                       | McCoy 5A + 10% FBS + 1% P/S + 10% Glycerol                     |
| MDAMB 435                  | DMEM + 10% FBS + 1% P/S + 10% Glycerol                         |
| MDAMB 435 Rho <sup>0</sup> | DMEM + 10% FBS + 1% P/S + 10% Glycerol + 50 µg/ml Uridine      |
| SW480                      | McCoy 5A + 10% FBS + 1% P/S + 10% Glycerol                     |
| C2C12                      | DMEM + 10% FBS + 1% P/S + 10% DMSO                             |

## Dose – response effect

### Materials

1. 24 well dishes (Nunc, ThermoFischer)
2. Resveratrol (Sigma) (Resv); Stock: 100 mM dissolved in DMSO
3. Sodium butyrate (NaBu); Stock: 1 M, dissolved in PBS
4. DMSO (Sigma)
5. PBSw/o
6. Sodium citrate (SC: 85mM sodium citrate, sterile)
7. PBS with  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$  (PBSw)
8. Counting solution (0.9% NaCl)

Note: Max 0,1% vol/vol of DMSO in the media

### WEEK 1 EXPERIMENT

#### Setup NaBu dose-response (week 1)

Sodium butyrate is produced by intestinal microbiota and act as a competitive histone deacetylase (HDAC) inhibitor. NaBu has a diverse range of effects on enterocytes.

This experiment is performed with both Caco-2 cells and DLD-1 cells.

### Procedure

1. Do not initiate the trypzination unless you are ready to seed for Exp 1, Exp 2, Exp 3 and Exp 4 and for continuous growing. **Calculate beforehand the number of cells needed to complete seeding for all setups.**
2. Trypsinize and count the cells as described on **page 19-23**. Record 1) counts, 2) cells/mL, 3) number of cells in total and 4) the passage number.
3. Make sure you have enough cells for the next passage, Exp 1, Exp 2, Exp 3 and Exp 4, before you proceed. Consult with your supervisor.
4. Seed (for both DLD-1 and Caco-2 in experiment 1) 25.000 cell/mL, 1.0 mL/well in 24-well plate in column 1-4, 20 mL is enough for four columns. The rest of the plate is left empty. One 24-well plate for DLD-1 and one 24-well plate for Caco-2.

5. Make sure to keep the rest of the trypsinized cells to seed for Exp 2, 3 and 4. Remember to resuspend cells from time to time, to ensure even pipetting of cells to all wells.
6. Let the seeded cells attach in the wells O/N.

**Time: 0 hour** (the following day, Tuesday)

7. Prepare dilutions of NaBu from 0.5-5 mM together with 0 mM (control with PBSw) in medium, at least 7 mL of each concentration. Keep the concentration of PBSw constant in each tube!
8. Replace the old medium with 1.0 mL of the NaBu-containing medium according to the scheme below.

| Col/Row  | 1                         | 2                  | 3                | 4                | 5            | 6 |
|----------|---------------------------|--------------------|------------------|------------------|--------------|---|
| <b>A</b> | <b>Control<br/>(PBSw)</b> | <b>0.5 mM NaBu</b> | <b>1 mM NaBu</b> | <b>5 mM NaBu</b> | <b>Empty</b> |   |
| <b>B</b> |                           |                    |                  |                  |              |   |
| <b>C</b> |                           |                    |                  |                  |              |   |
| <b>D</b> |                           |                    |                  |                  |              |   |

**Time: 48 hr** (Thursday)

9. Acquire phase contrast pictures (Leica DM IRB, EVOS M5000 both in room 28A2-37). One picture in the center of each well to obtain optimal phase contrast, use 20 x magnification. Save on the computer in .tiff format. Note: Because you want to have cells surviving in good health for the Coulter counting step below, it is best to be as quick as possible when acquiring images. Generally, only remove one 24-well plate at a time from the incubator.
10. Wash the cells 2x with 1 mL PBSw/o or 1 mL SC, add 0.2 mL trypsin, incubate 20 min, add thereafter 0.8 mL PBSw/o.
11. Resuspend on ice, which will reduce cell aggregation.
12. Transfer 0.8 mL cell suspension to 10 mL counting solution (0.9% saline).
13. Estimate the cell number using the Coulter counter (**page 20-21**). Count the particle size above 9  $\mu\text{m}$ .

14. For calculating the cells/mL, remember the dilution factor (800 $\mu$ L/10800 $\mu$ L!) and a total volume of 0.5 mL is used for the counting.
15. What to do with the data? To calculate cell diameter (a proxy for cell size) see **page 34** (“Refinement of the counting and determination of cell diameter”). To measure area from images, see **page 34-36**.
16. The aim for the week 1 experiment is also to try out the assay with few concentrations of NaBu in order to learn the technique prior to setting up a large experiment (Week 2).

## **WEEK 2 EXPERIMENT**

### **Setup NaBu and Resveratrol dose response (week 2)**

Based on the results from Week 1, select the concentration of NaBu to use together with increasing concentrations of Resv: Use a concentration of NaBu, which has have significant effects on cell counts as well as on cell diameter, but not reducing the cell number completely. Consult with instructors.

Team 1 + 3 will use DLD-1 cells for Week 2 setup

Team 2 + 4 will use Caco-2 cells for Week 2 setup

### **Procedure**

1. Do not initiate unless you are ready to seed for Exp 1, 2, 3 and 4. Calculate beforehand the number of cells needed to complete seeding for all setups.
2. Trypsinize and count the cells as on **page 19-24**. Record 1) counts, 2) cells/mL, 3) number of cells in total and 4) passage number.
3. Seed two 24-well plates with DLD-1 **or** two 24-well plates with Caco-2 cells: 25.000 cell/mL, 1.0 mL/well in 24-well plate, 55 mL cell suspension is enough for two plates.
4. Make sure to keep the rest of the trypsinized cells to seed for Exp 2, and 3.
5. Let the seeded cells attach O/N.

**Time: 0 hour** (the following day)

6. Prepare dilutions of Resv according to the matrix below. For Resv, prepare medium with 2.5, 5, 10, 20, and 40  $\mu\text{M}$  in combination with one concentration of NaBu selected above, as well as medium with no test compound but only solvents (control) in medium, at least 5 mL of each concentration. Keep the concentration of PBS and DMSO constant between conditions!

Hint: Prepare solutions with double concentrations of the highest level of Res and NaBu.

7. Replace the old medium with 1.0 mL of the Resv and NaBu-containing medium according to the scheme below. The two dishes are treated exactly in the same way:

|   | 1       | 2                                | 3                              | 4                               | 5                               | 6                               |
|---|---------|----------------------------------|--------------------------------|---------------------------------|---------------------------------|---------------------------------|
| A | Control | 2.5 $\mu\text{M}$ Resv           | 5 $\mu\text{M}$ Resv           | 10 $\mu\text{M}$ Resv           | 20 $\mu\text{M}$ Resv           | 40 $\mu\text{M}$ Resv           |
| B |         |                                  |                                |                                 |                                 |                                 |
| C | NaBu    | 2.5 $\mu\text{M}$ Resv<br>+ NaBu | 5 $\mu\text{M}$ Resv +<br>NaBu | 10 $\mu\text{M}$ Resv<br>+ NaBu | 20 $\mu\text{M}$ Resv<br>+ NaBu | 40 $\mu\text{M}$ Resv<br>+ NaBu |
| D |         |                                  |                                |                                 |                                 |                                 |

**Time: 48 hr**

1. Acquire data exactly as for WEEK 1.
2. Additionally, prepare a single 40x phase contrast picture of each relevant exposures: Control, Resv, NaBu and the combination.
3. To calculate cell diameter based on coulter counter data and area (proxies for cell size) based on microscope images. See 'Refinement of the counting and determination of cell diameter'. Estimate dose-response of Resv for the two different cell lines.
4. The aim is to identify the combinatory effect of NaBu and Resv. Moreover, we will determine if this response is similar in DLD-1 and Caco-2 cells. Compare your findings to those from the other groups to determine this.

## For the report – Exercise 1

### Sub cultivation, and growth rate

- Include your laboratory notebook (scan the pages related to experiment 1)
- Briefly, describe challenges in generating reproducible data using cells growing in culture.
- How were the on-going cell cultures growing? Which challenges did you experienced?
- Construct a table showing the total cell counts obtained from the Coulter counter and the hemocytometer (use your sub cultivation data) together with the number of cells you seeded 3-4 days in advance.
- Determine the growth rate. Use the following equation

$$(t_2 - t_1) \times \left( \frac{\text{Log}(2)}{\text{Log}\left(\frac{Q_2}{Q_1}\right)} \right)$$

to estimate a provisional cell division time based on the cell counts from your maintenance cultures, where  $t_1$  and  $t_2$  are the times of cell counting, and  $Q_1$  and  $Q_2$  the respective total cell numbers.

- How successful was the freezing-thawing procedure?

### Specifically, for the Dose-response experiments:

- Prepare a dose-response curve after exposing cells to the cytostatic compound sodium butyrate. Prepare curves for Caco-2 and DLD-1 setups (week 1). Be aware that the cell counts from the coulter counter need adjustments. Look at the histogram of coulter counting.
- Prepare a dose-response curve after exposing cells to the cytostatic compound Resv. Prepare curves for Resv stimulation and for Resv stimulation in combination with NaBu (week 2).
- As Resv changes cell size, you should determine the cell areas (from images) and the cell diameter (from coulter counter) in response to this. It is possible that NaBu



also changes cell size, and to determine this, you should determine the cell area and the cell diameter in response to NaBu and in response to the Resv and NaBu combinatory treatment.

- Describe and discuss the dose-response effect of Resv and NaBu on cell proliferation and cell size,
  1. by graphic presentations and by representative images of Caco-2 or DLD-1 cell lines, as appropriate for your group.
  2. by a relevant text.
- Describe factors that are determining the rate of cell proliferation and cell growth.
- What are the advantages or limitations of using Coulter counting compared with other cell proliferation/viability assays (e.g. Hoechst staining, Resazurin metabolism, Neutral Red assay or XTT assay, exercise 2)
- Compare the effect of Resv on cell size when measuring the area and cell diameter. Why do you obtain different numbers?
- Which cellular structures can you identify using phase contrast microscopy (use 40x magnification and likely adjust the focus level)?

## Appendix 1.1 Determination of cell area, refinement of the counting and determination of cell diameter

The MultiSizer 4e software, which is used to count particles and store the counts, has several additional features:

1. You may remove the particles less than a given size e.g. 10.5  $\mu\text{m}$  in DLD-1 cells, which may be cell debris etc. Remember to multiply the count with dilution factor and factor 2 if you are counting 0.5 mL (coulter counter setup).
2. You may also make a rough estimate of the mean size of the counted cells.
3. The numbers may be exported to Excel to perform the subtraction of the numbers of non-cell particles. For the export of cell counts to an Excel format, see “Determination of cell diameter using FitterBJ03“. We have made an Excel file to import the numbers into MultiSizer Import template 2023.xls. (Moodle folder).
4. Using the same export files from the MultiSizer software you may determine the mean diameter of the cells using software developed by Benjamin Toft, Århus (FitterBJ03).
5. Distribution curves may be drawn to see the effect of treatment on cell diameter.
6. Install the MultiSizer 4e software (Moodle) on your computer (Windows). The FitterBJ03 is a Java based software, so Java need to be installed on your computer as well.

### Determination of cell area

#### Acquire pictures of the cells

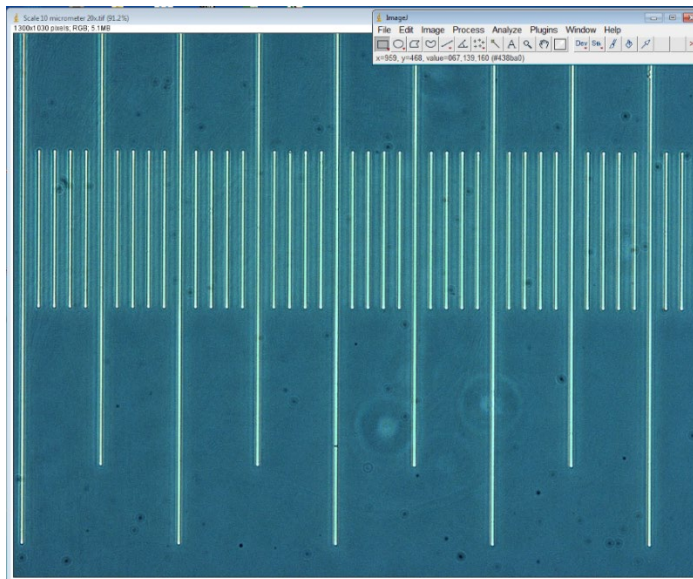
1. Suitable phase contrast pictures to determine cell area have:
  - a. Optimal contrast (picture is taken from the center of the well – bigger dishes (etc. 3 cm) reduce this problem).
  - b. The picture should cover at least 10 cells with an uncovered edge.
  - c. Make pictures with 20x magnification.

2. The pictures are made with the LCD camera placed on the LEICA DM IRB microscope (28A.2-37).
  - a. Alternative use the EVOS M5000 (28A.2-37).
3. Acquire one picture of each well.

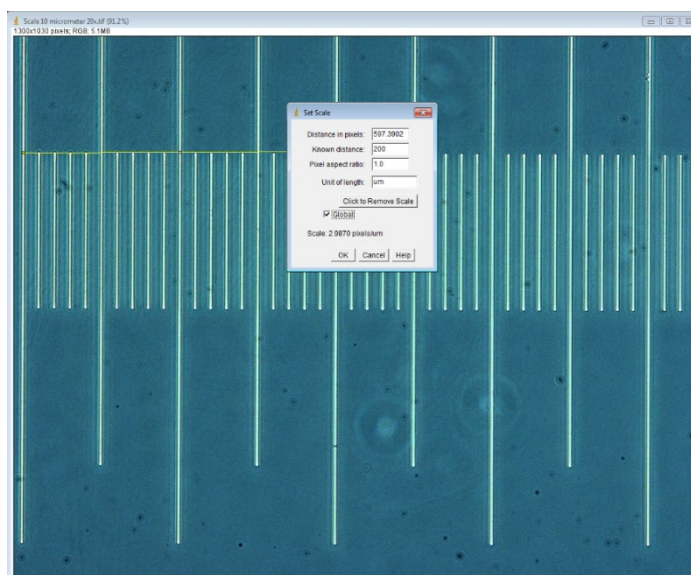
### Area determination (using Fiji ImageJ software)

4. Start the Fiji ImageJ software (download from Moodle or <https://imagej.net/Fiji/Downloads>) for windows.

5. Adjust the scaling
  - a. Use the file scale 10 micrometer 20x.tif (Moodle)
  - b. Use **STRAIGHT LINE SELECTION** and indicate the length of 200  $\mu\text{m}$  on the picture:
    - i. The distance between each grid is 10  $\mu\text{m}$ , and 100  $\mu\text{m}$  is the distance between 10 grids!

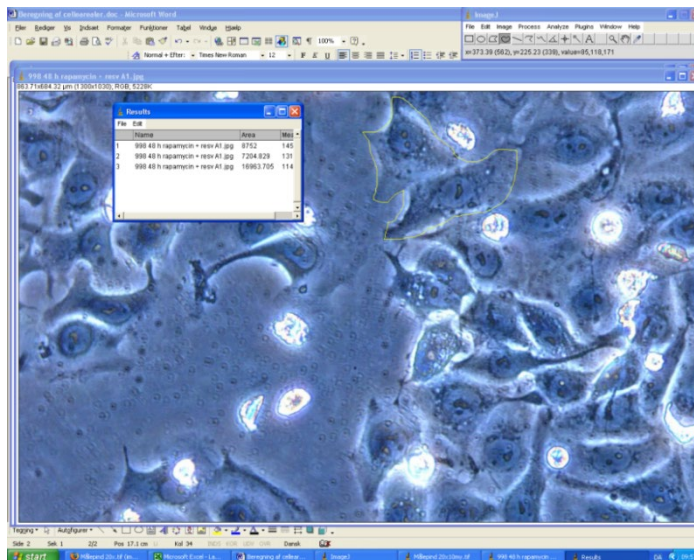


- c. Find **SET SCALE** under **ANALYZE**, and adjust:
    - i. **KNOWN DISTANCE:** 200
    - ii. **UNIT OF LENGTH:**  $\mu\text{m}$  (you can't write  $\mu$ !)
    - iii. Mark **GLOBAL**.
    - iv. **OK**
  - d. To check the scaling:
    - i. Use the polygon icon and draw a figure with a known area on the scaling picture
    - ii. Press **MEASURE** under **ANALYZE** (or use Ctrl M).
    - iii. A new window shows Results
6. Area determination of cells
  - a. Open the relevant file



- b. Use **FREEHAND** selection
- c. Draw around the relevant cells:

- i. Include several cells in one outline. Include a number of cells (5 or more) which are placed to the edge of cell island and include the first and second row of cells from the edge. Remember (and write down) how many cells you include.
  - ii. Press **MEASURE** under **ANALYZE** (Ctrl M).
  - iii. Repeat the determination until you have 10-15 cells from each well.
  - iv. Note: You need some practice to sketch the girth of the cells. If you are not satisfied, you may start from another point on the cell and make a new round. The previous areas obtained using **MEASURE** function is still present. In a case you want to redo also the **MEASURE**, then erase the value by clicking the right side mouse-button and select **CLEAR**.
  - v. Note: The objectivity is a problem here because you may select cells with the size you want: big or small. Make your own rules saying e.g. that you are counting three cells from the top and beginning the area determination from this and further five cells down. Another (better) way is to do the area determination blinded so the person making the measurements does not know the code of the sample.
7. Results:
- a. The area data may be imported to Excel.
  - b. In the Results window: **SELECT ALL**, and **COPY**.
  - c. In Excel: **PASTE**.



## Export of data from MultiSizer 4e and determination of cell diameter using FitterBJ03

This instruction describes the use of FitterBJ program. If you identify problems, please give me a note.

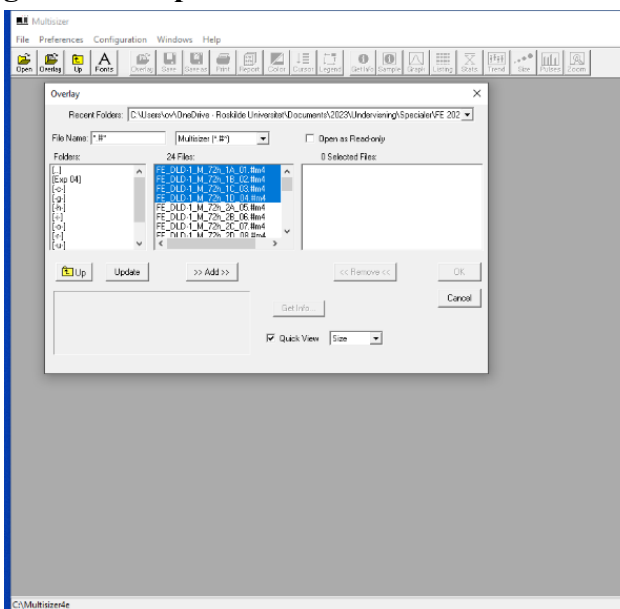
### 1. Installation of FitterBJ:

- a. The program is developed by Benjamin Jakobsen, Århus
- b. A copy of the program is found on Moodle

- c. Copy the program to the desktop of your computer (windows), no installation is needed.
- d. The data are exported from the MultiSizer program in a csv-format.

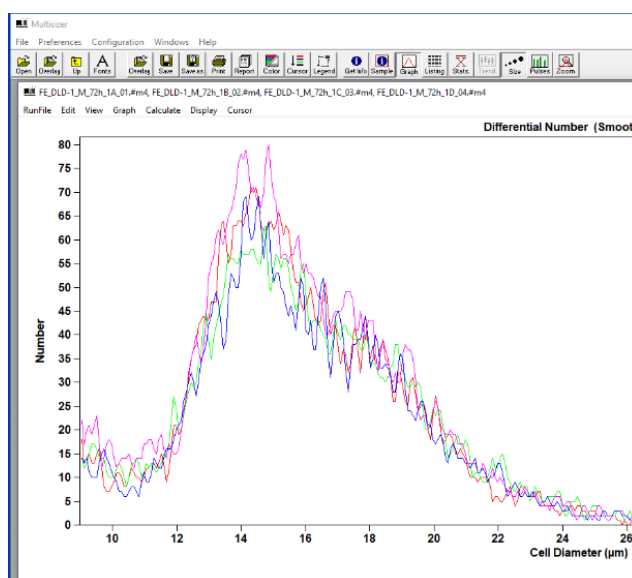
## 2. Open files in the MultiSizer program and export to FitterBJ:

- a. Open the **OVERLAY FILES-**  
ikon AND apply the files you  
want to measure.

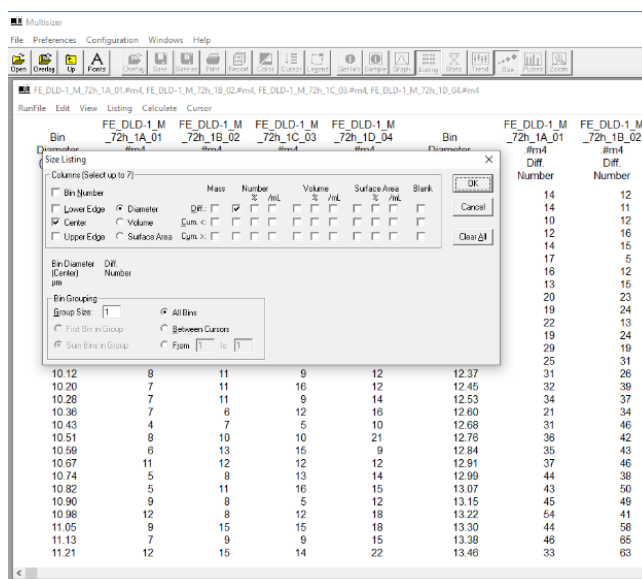


- b. The following screen is seen:

- c. Check that the row sequence is correct: A1, B1, C1, D1, A2, B2, C2 etc. If not, use **Arrange runs** under **Cursor**.



- d. Use **VIEW** in the lowest command-line choose **LISTING**:
- e. Choose **COLUMN FORMAT** under **LISTING**.
- f. Check that all  $\sqrt{\phantom{x}}$ -marks are in the following order (see the figure):
  - i. Center
  - ii. Diameter
  - iii. Number diff.



- g. Press **EXPORT DATA** under **RUN FILE**:

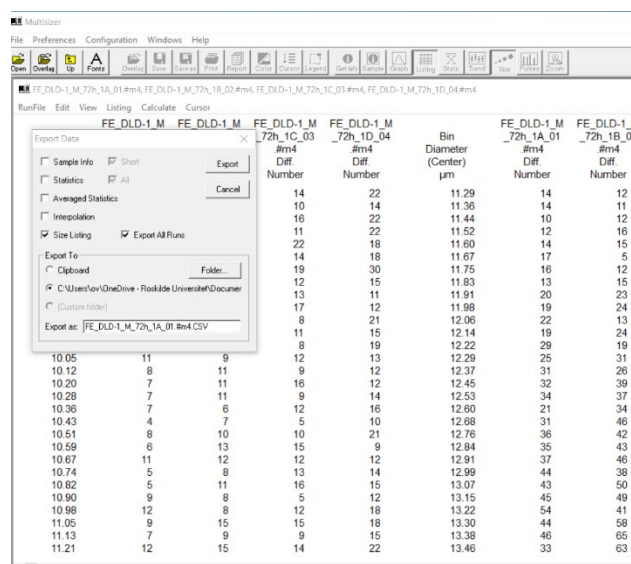
- h. Remove  $\sqrt{\phantom{x}}$ -marking in **SAMPLE INFO** and **STATISTICS**

- i. Add  $\sqrt{\phantom{x}}$ -marking in **SIZE LISTING** and **EXPORT ALL RUNS**

- j. Give the right file name (which will obtain the extension -.xls) in **EXPORT AS**.

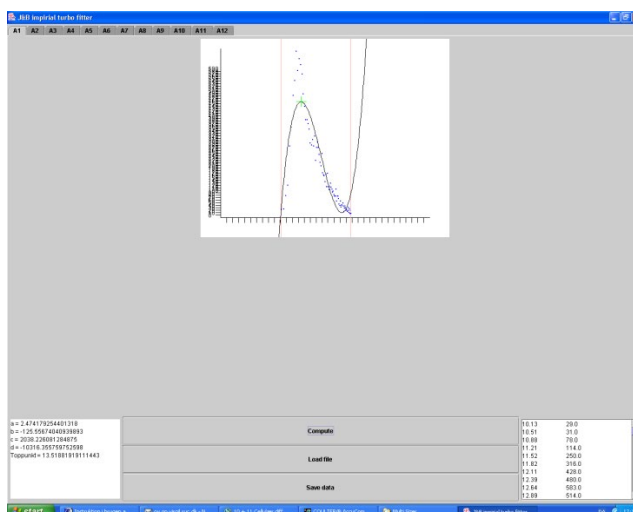
- k. Press **EXPORT**.

- l. The program will make an export file (name.xls), which may be read by the fitter program. The file will be located in the same folder as the original Multi Sizer data file.



### 3. Run FitterBJ:

- a. Activate the icon for the FitterBJ program.
- b. **OPEN** window is popping up. Find the file made by the Multi Sizer export comando.
- c. Load the file by pressing **OPEN**.
- d. The new window contains three areas (figure):
  - i. Histogram and the fitting (top middle)
  - ii. Calculated constants for the identified fitted curve (below left). These numbers may be saved by the **SAVE** function.
  - iii. List of numbers (diameter versus numbers) (below right)
- e. The number of data set in the Multi Sizer export file will add the same number of pages in the fitter program, named A1, A2, A3 etc.
- f. Start with A1. Click on the first data point (left mouse key) where the fitting should begin (e.g. the lowest point). Following, click on the last point of fitting (right mouse key), (first time 0 is present). Then press **compute**. Follow with page A2.
- g. All fitted data may be saved by **Save**. These data may be collected in an excel-spread sheet. Or, write down the top point in each calculation
- h. **LOAD** will load a new Multi Sizer export file.



## Appendix 1.2 Calculation of volumes of solution for dose-response tests

Calculations of the volumes of cell solutions or specific doses of compounds are often a challenge for students. Below are some hints.

1. When you want to compare effects, everything should be kept constant beside the substance or treatment you want to measure. So, the volume of vehicle(s) (e.g. DMSO or PBS) should be the same in all the groups you want to compare.
2. Keep the volumes of the expensive compounds low, because some of the compounds are .... expensive or you may only have few mg!
3. Do not take out less than 1  $\mu\text{L}$ . For you as non-trained the uncertainty in pipetting such small volumes is very high.
4. The total amount of solvents like DMSO should not exceed 0.1 % (vol/vol).

Example:

You are exposing your cells to 0.1 – 10 mM NaBu, and you have a 1 M NaBu stock solution in PBSw.

- How much volume are you going to use? In the 24 well setup, you have four treatments/ group, and 1 mL / well, so prepare 5 mL of each concentration.
- 10 mM is the highest exposure of NaBu. How much should you take out of your stock solution to get 5 mL medium with 5 mM NaBu?
  - This may be done in different way, but I always ask: How many times do I need to dilute my stock solution to get this concentration?
  - $1\text{ M} \rightarrow 5\text{ mM}$ :  $1000\text{ mM} / 5\text{ mM} = 200\text{x}$  dilution.
  - Then, what is 1/200 part of my final volume?  $10\text{ mL} = 10000\text{ }\mu\text{L} \Rightarrow 10000\text{ }\mu\text{L} / 200 = 50\text{ }\mu\text{L}$ .
  - So, if you take out 50  $\mu\text{L}$  of the 1 M NaBu solution in 10 mL medium, you will obtain a 5 mM solution.
- Next, prepare the control sample
  - Add the same volume of the vehicle as for the highest concentration.
  - $50\text{ }\mu\text{L} / 10\text{ mL} = 5\text{ }\mu\text{L} / \text{mL}$ .

As you will see below, you may need more than 10 mL of your control samples, so prepare 15 mL in total and add 75  $\mu\text{L}$  PBSw.

Your highest concentration and control sample do now contain the same amount of vehicle PBSw.

- and mixing these two will keep the concentration of the vehicle constant and only the concentration of NaBu will be changed.
- 1 mM NaBu: mix 6 mL of control with 1.5 mL of 10 mM NaBu solution.
- 0.5 mM NaBu: mix 2.5 mL of control with 2.5 mL of 1 mM NaBu solution.

In week 2 we have solutions with two different solvents (DMSO and PBS). Both solvents need to be the same in all treatments!

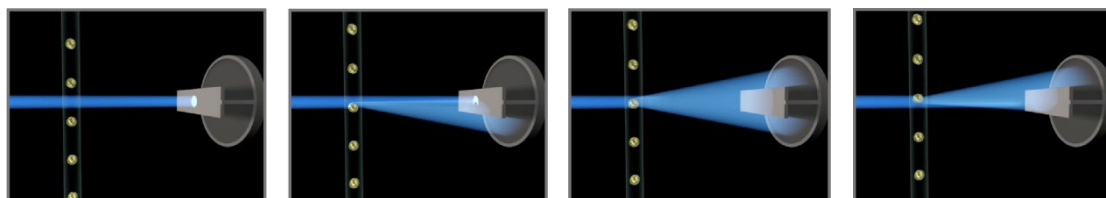


## Exercise 2 – Cell cycle and viability analysis

In this exercise, you will learn how to estimate modulation of cell cycle and cell proliferation following exposure to NaBu and Resv. You will become more familiar with the FACS and the plate reader (for the viability assay) and learn to set up in different well formats.

### Introduction to flow cytometry

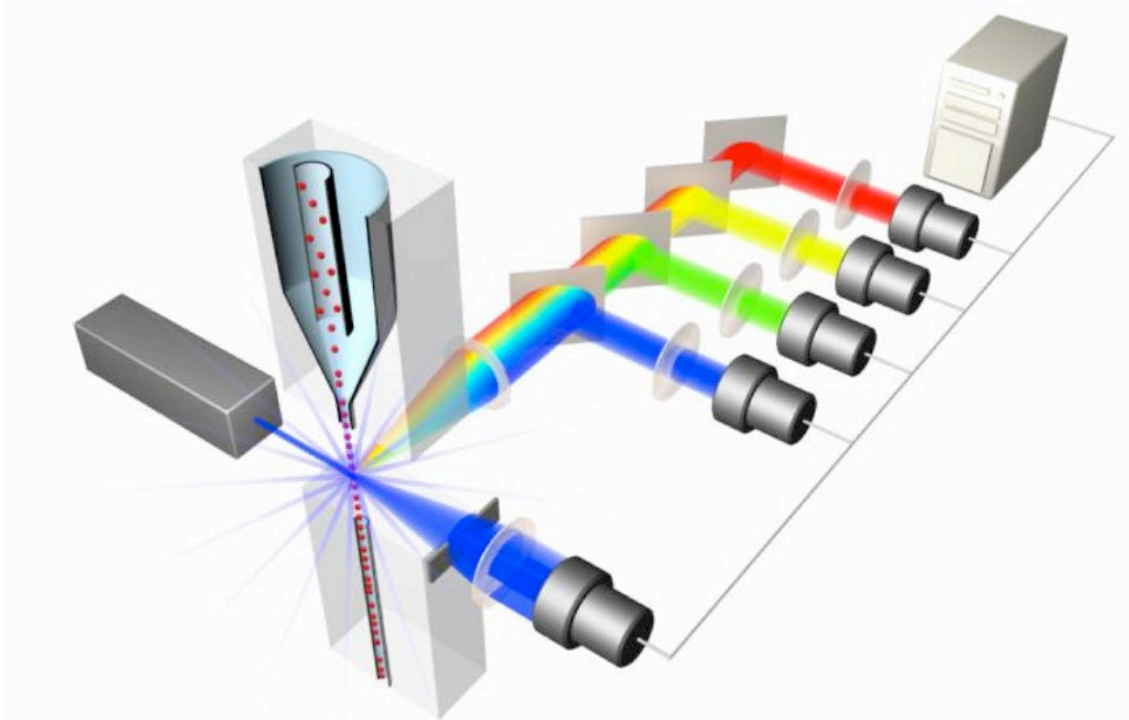
Flow cytometry is a technology that allows a single particle to be measured for a variety of characteristics. The particles measured may be cells, but also chromosomes, latex beads or any other particle that can be suspended in fluid. The flow cytometer performs this analysis by passing thousands of particles per second through a laser beam and capturing the light that emerges from each cell as it passes through. The data gathered can be analyzed statistically by flow cytometry software to report cellular characteristics such as size, complexity, phenotype, and health.



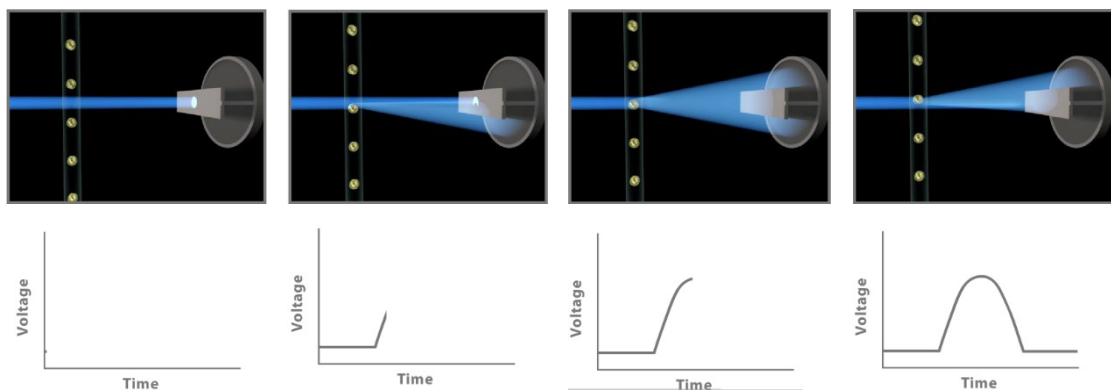
Each suspended particle passing through the beam scatters the light in some way, and fluorescent chemicals found in the particle or attached to the particle may be excited into emitting light at a lower frequency than the light source. This combination of scattered and fluorescent light is registered by different detectors, and by analyzing fluctuations in brightness at each detector, it is possible to obtain information about the physical and chemical structure of each individual particle.

#### Detectors

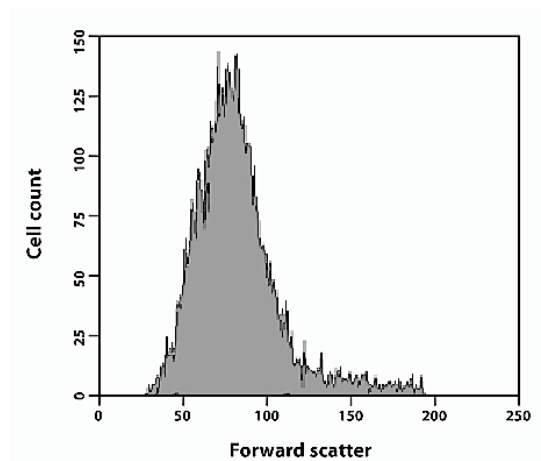
The detectors responsible for extracting information from the passing particles are primarily the Forward Scatter (FSC), the Side Scatter (SSC), and a number of photomultipliers.



**Forward scatter (FSC):** The detector measures the size of the particles. It does so by measuring the intensity of the light scattered by the particle



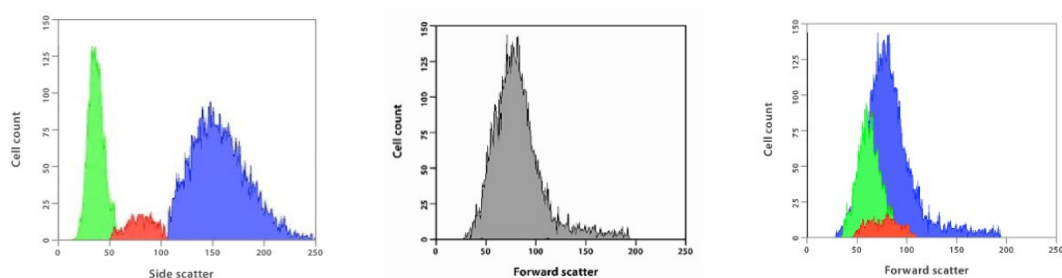
The intensity is measured in volts and the magnitude of the voltage pulse recorded for each particle is proportional to the size of the particle. When compiling the intensity of each particle measured in a histogram, it is possible to get a picture of the size distribution of the population of particles.



**Histogram of voltages recorded with FSC:** Smaller particles appear toward the left and larger particles appear toward the right. A histogram of forward-scatter data is a graphical representation of the size distribution within the population.

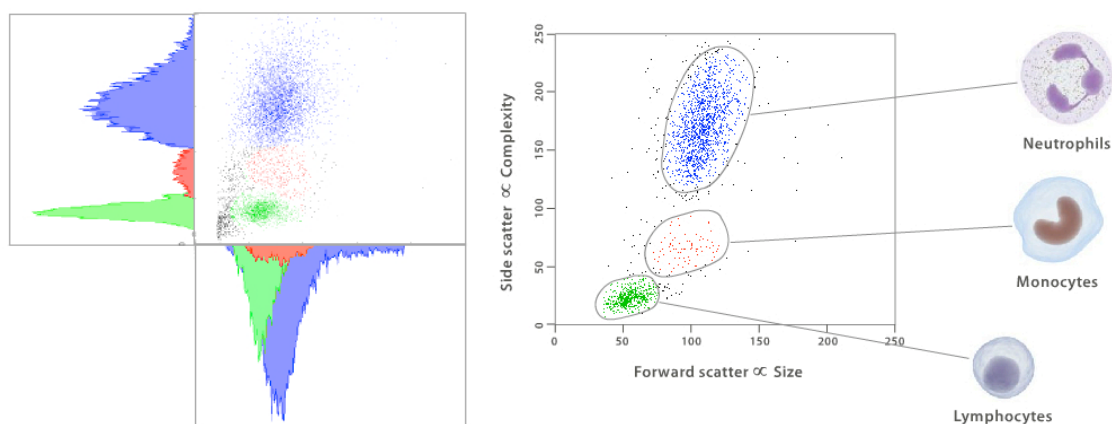
It is very important to notice, that the size measurements not necessarily are identical to values obtained by Coulter counting. The signal from the FSC reflects the size but also the refractive index of the particle.

**Side scatter (SSC):** By combining FSC with data from SSC, much more data is available from each particle. The SSC detects light that are scattered to an angle of  $90^\circ$  from the illuminating beam. The rougher or more irregular or granular a particle is, the more it will scatter the illuminating beam to the side. The signal from the SSC is therefore related to the particles surface texture and internal structure as well as to its size and shape.



**Left histogram:** The data obtained from SSC can be plotted in a histogram as the data from the FSC was. It is very evident that there exist three different groups of particles in the population. **Middle histogram:** FSC data of the population. **Right histogram:** By integrating the data from the SSC in the FSC histogram from before, it is evident that we get a more detailed description of the characteristics of the particles.

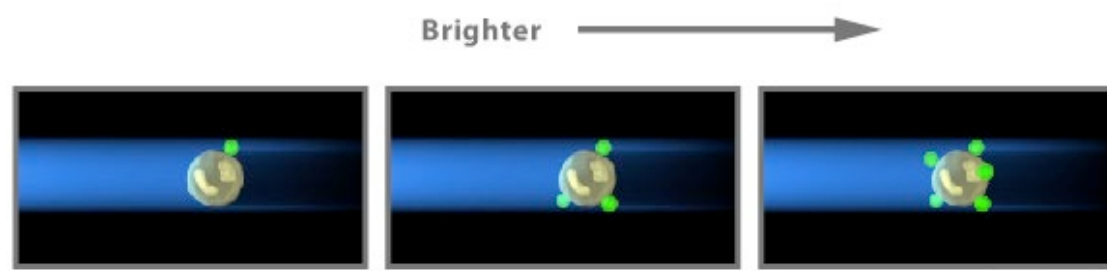
We now can classify the population of particles into groups related to the size and the complexity of the particles. To give a better overview of the data, the FSC and SSC histograms are often merged into a 2d histogram.



By using the 2d histogram it is very easy to see the different types of particles. In this example, a blood sample is divided into neutrophils, monocytes, and lymphocytes according to cellular size and complexity.

**Photomultipliers:** We now have a tool for differentiating the population of measured particles by size and complexity. By using additional photomultipliers, it is possible to measure any light emitted from the particle. The emitted light can come from the particle

itself or more useful, from a fluorochrome that has attached itself to the particle. When a laser beam of the right wavelength strikes the fluorophore, a fluorescent signal is emitted and, through the photomultipliers, detected by the flow cytometer. For cells, the fluorochrome can be delivered as a fluorochrome antibody or a compound that has a high specificity for some part of the cell. Propidium iodide intercalates strongly with DNA for example.



The emitted light from the fluorochromes is detected by an appropriate photomultiplier as a voltage pulse proportional to the amount of fluorescence emitted.

Flow cytometry is a use full tool for measurement of cell cycle phases. The measurement is based on the fact that the genome is divided in the S-phase. In the G<sub>2</sub>/M-phase the amount of nuclear genomic DNA is twice that of the G<sub>1</sub>-phase. By labeling genomic DNA with **propidium iodide**, one is able to discriminate whether individual cells are in G<sub>0</sub>/G<sub>1</sub>, S or G<sub>2</sub>/M-phase based on their fluorescence intensity.

Apoptosis may also be measured by flow cytometry (not in the present experiment). Cell death may happen via two distinct routes: Necrosis or apoptosis. Necrosis is the pathological result of cells being exposed to a severe physical or chemical insult. Apoptosis, or programmed cell death is a more physiological process, where unwanted cells are eliminated during i.e. development or through a normal biological process. There are a number of features which set apoptosis apart from necrosis. Apoptosis is a distinctly regulated process, where a number of cellular proteases are activated (caspases) and help the cell to commit suicide, it requires ATP, mitochondria release cytochrome C into the cytoplasm and DNA is fragmented into nucleosomal fragments.

Alterations in membrane asymmetry is a step of apoptosis and measured in the **Annexin V assay**, which allows identification of cell surface changes that occur early during the

apoptotic process. Early in the apoptosis process, phosphatidylserine (PS) becomes exposed on the cell surface by flipping from the inner to the outer leaflet of the cytoplasmic membrane. The binding of Annexin V to phosphatidylserine is identifying membrane changes associated with apoptosis, using either flow cytometric or *in situ* labeling methods.

## Introduction to Hoechst 33342 & Alamar blue viability assays

Determination of cell number, cell proliferation and / or cell viability is crucial in the understanding of the cell biological effect of a given treatment. The number of available assays is tremendous. Table 2.1 only focus on assays available in a 96 well setup, which therefore have a significant power of screening.

| <b>Table 2.1</b>                             |                                                                                                               |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Assay</b>                                 | <b>Principle</b>                                                                                              |
| 5-bromo-2'-deoxyuridine (BrdU) incorporation | Measuring new DNA synthesis                                                                                   |
| <sup>3</sup> H-thymidine incorporation       | Measuring new DNA synthesis                                                                                   |
| Hoechst 33342                                | Measuring total DNA                                                                                           |
| MTT (as well for XTT, MTS and WST-1)         | Reduced by metabolically active cells to a blue-purple formazan product                                       |
| Alamar Blue (Resazurin metabolism)           | Reduction of the oxidized blue dye, with low fluorescence, to its pink, fluorescent resorufin product         |
| Neutral Red                                  | Fluorescent compound, absorbed into lysosomes                                                                 |
| Sulforhodamine B                             | Form an electrostatic complex with the basic amino acid residues of proteins under moderately acid conditions |
| Trypan blue                                  | Uptake of a membrane impermeable substance                                                                    |
| Bioluminescent ATP assays                    | Measuring ATP concentrations firefly luciferase                                                               |

The MTT assay is often used as a gold standard that new viability/cytotoxicity assay methods are compared with, but the various assays have different benefits and drawbacks. E.g. MTT is reduced by glutathione S-transferase (GST), why MTT not always

give a reliable cell viability estimate when the compounds under test affect the GST activity. Some compounds, such as flavonoids, have the ability to reduce MTT nonspecifically in the absence of cells. This may mask a cytotoxic/cytostatic effect. Several of the dyes (including Hoechst 33342 and Sulforhodamine B) may be excluded by cellular drug transporters and therefore give an underestimation of the cytotoxic/cytostatic effect.

## Experimental procedure

The aims of the current experiment are 1) to introduce the flow cytometer and Hoechst 33342 and Alamar Blue viability assays; and 2) to use the assays to estimate the influence of NaBu and Resv on cell cycle phase distribution and cell viability. As for Exp 1 there will be two slightly different setups: one for WEEK 1 and one for WEEK 2.

**For each week there are 2 parts: 1) Flow cytometry for Cell cycle and 2) Viability test using the Hoechst 33342 and Alamar Blue assay**

Team 1 + 3 will use DLD-1 cells for Week 1 and Week 2 setups for cell cycle analysis. Team 2 + 4 will use Caco-2 cells for Week 1 and Week 2 setups for cell cycle analysis. All teams will test both DLD-1 and Caco-2 cells using the Hoechst 33342 and Alamar Blue viability assays.

### WEEK 1 EXPERIMENT

## Experiment 2A – Cell cycle analysis

### Materials

- Propidium Iodide (0.4 mg/ mL)
- RNase A (1 mg/ mL)
- PBS with  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$  (PBSw)
- Sodium butyrate, NaBu stock (1 M)
- PBS-EB: PBS w/o + 5 mM EDTA + 0.5% BSA
- 70% ethanol

### Day 1: Seeding of DLD-1 or Caco-2 cells

1. Use cells trypsinized in Exp 1 to seed:



2. For Flow cytometry: 2 T75 flasks per cell line: two flasks with 700.000 cells (for 24 hrs treatment) in a total of 12 mL medium each.
  - Week 2: 3 T-75 flasks

### **Day 2: Treatment**

3. Expose the cells to PBSw or 10 mM NaBu.
  - Week 2: PBSw, 20  $\mu$ M Resv or 10 mM NaBu

### **Day 3 (24 hrs): Preparation of cell samples for Flow cytometry)**

1. Trypsinize cells. Resuspend cells in PBS-EB (PBS w/o + 5 mM EDTA + 0.5% BSA) and pellet cells by centrifugation for 5 min at appr. 300g, about 1800 rpm. (Note: If you want to analyze apoptotic cells as well, remember to collect the cells floating in the media – but we do not include these cells for this experiment). Discard supernatant.
2. Wash cells in appr. 10 mL PBS-EB
3. Spin down as before. Discard supernatant. Resuspend cells in 3-5 mL PBS-EB.
4. Count the cells. Transfer 500.000 cells to a FACS tube (BD, Falcon 5 mL polystyrene round-bottom tube, cat # 35 2058). Add carefully PBS-EB until the total volume is appr. 3 mL. Spin down the cells and discard supernatant. Make up to 3 of these FACS tubes for each cell line.
5. Resuspend cells in exactly 200  $\mu$ L PBS-EB. Add slowly 2 mL ice-cold 70% ethanol while whirly mixing (IMPORTANT). Incubate on ice for at least 30 min. The samples should be kept at  $-20^{\circ}\text{C}$  at this stage.

### **Staining the cells for Flow cytometry**

1. Centrifuge the cells at 300g and  $4^{\circ}\text{C}$  for 5 min. Discard supernatant.
2. Resuspend cells in 2-3 mL PBSw. Spin as before. Discard supernatant.
3. Resuspend cells in 400  $\mu$ L PBSw. Add 50  $\mu$ L RNase (1 mg/ mL) and 50  $\mu$ L propidium iodide (0.4 mg/ mL). Incubate at RT for 30 min in the dark.
4. Centrifuge the cells at 300g and  $4^{\circ}\text{C}$  for 5 min. Discard supernatant.
5. Resuspend cells in 0,5 mL PBSw

6. Analyze by flow cytometry.

It is a good idea to set up extra tubes for each cell sample (make three tubes per sample), so you don't have to repeat the whole experiment if a tube is lost.

Also, remember to make some tubes (i.e. 3 tubes) with untreated cells to test for background fluorescence and to use for optimizing the instrument settings before doing the FACS analysis.

**Note:** Propidium iodide is toxic. Wear gloves and exercise caution when handling propidium iodide.

**Note:** Propidium iodide is light sensitive and should be kept in the dark as much as possible during labeling and handling. Keep tubes covered with aluminum foil or place in a drawer during incubation steps (easiest). Keep cells in the dark after labeling. Brief exposure to light during pipetting is acceptable.

**Flow cytometry**

- Collect data on flow cytometer: FACS Celesta: FSC height, FL2 – PI height. See appendix 2.1
- Use linear scaling. Use one sample to adjust the flow cytometer so that G1 is set to 100 and G2 to 200 in the linear scaling.
- Because we only use one-color flow cytometry we do not need to adjust for overflow between fluorescent channels.

**Analysis of results**

- Present data from each sample in an Analysis Dot Plot
- Draw a region around the cells to exclude cells that have not been trypsinized completely (doublets)
- Use the same Region on all Dot Plots (copy between plots) and gate this region. Now only cells inside the region will be analyzed
- Save the Cell Quest document.

- Open the Cell Cycle Analysis program and import sample file.
- Use standard settings to calculate cell cycle distributions (percent of cells in each phase).

## Experiment 2B – Hoechst 33342 / Alamar Blue 96 well assays

The principle of this assay is based on the fluorescence of the DNA binding dye Hoechst 33342. By adding the cell permeable Hoechst dye, we can quantify the number of nuclei, and hence number of cells, in a 96 well plate format.

In the Alamar Blue assay, Resazurin is converted primarily by dehydrogenases which turn faint blue into pink fluorescent, but it is not strictly clear which enzymes are responsible .

### Materials

1. White 96 well dishes, to be seeded with DLD-1 and Caco-2 cells (Nunc, ThermoFischer)
2. Growth media, complete
3. Resveratrol (100 mM)
4. Sodium butyrate, NaBu stock (1 M)
5. Hoechst 33342 (Stock conc: 10 mg/mL (MiliQ H<sub>2</sub>O))
6. Resazurin (Stock conc: 10 mg/mL (MiliQ H<sub>2</sub>O))
7. Phosphate-buffered saline with Ca<sup>2+</sup> and Mg<sup>2+</sup> (PBSw)
8. Microplate reader (Synergy HT BioTek)

### Hoechst plate reader-assay

#### Day 1: Seeding of DLD-1 and Caco-2 cells

- Use cells trypsinized in Exp 1 to seed.

- Seed column 2-6 of a 96 well plate with DLD-1 cells and column 7-11 with Caco-2 cells: 10.000 cells per well. Seed in a volume of 200  $\mu$ L. 200  $\mu$ L medium without cells is added to column 1 and 12.

#### Day 2: Treatment with NaBu

- Use the concentrations of NaBu as indicated in the diagram below. Add a volume of 200  $\mu$ L for treatments (see Figure 2.1).

|   | 1                        | 2            | 3            | 4           | 5         | 6         | 7            | 8            | 9           | 10        | 11        | 12                       |
|---|--------------------------|--------------|--------------|-------------|-----------|-----------|--------------|--------------|-------------|-----------|-----------|--------------------------|
| A | No cells – media<br>only | Control -PBS | Control -PBS | NaBu 0.5 mM | NaBu 1 mM | NaBu 5 mM | Control -PBS | Control -PBS | NaBu 0.5 mM | NaBu 1 mM | NaBu 5 mM | No cells – media<br>only |
| B |                          |              |              |             |           |           |              |              |             |           |           |                          |
| C |                          |              |              |             |           |           |              |              |             |           |           |                          |
| D |                          |              |              |             |           |           |              |              |             |           |           |                          |
| E |                          |              |              |             |           |           |              |              |             |           |           |                          |
| F |                          |              |              |             |           |           |              |              |             |           |           |                          |
| G |                          |              |              |             |           |           |              |              |             |           |           |                          |
| H |                          |              |              |             |           |           |              |              |             |           |           |                          |
|   | DLD-1 cells              |              |              |             |           |           | Caco-2 cells |              |             |           |           |                          |

Figure 2.1

#### Day 4 (Thursday):

- Prepare growing medium containing Hoechst 33342 (5  $\mu$ g/mL) and resazurin (25  $\mu$ g/mL) by adding Hoechst 33342 and resazurin stock solutions into complete growth medium for each cell line. Use 100 $\mu$ L/well.
- Incubate for 30 min at 37°C in a CO<sub>2</sub> incubator.
- Read the resazurin fluorescence immediately:
  - Synergy HT, BIOTEK plate reader (28A.2-21)
  - Excitation 530/25 nm (Filter wheel D) and
  - Emission 590/20 nm (Filter wheel A), respectively. Gain: 45
- Wash three times with 100 $\mu$ L PBSw. Add 100 $\mu$ L PBSw as the last step.
- Read the Hoechst fluorescence
  - Synergy HT BioTek plate reader (28A.2-21)
  - Excitation 360/40 nm (Filter wheel D)
  - Emission 460/40 nm (Filter wheel A), Gain: 60.
- Calculate the effect of NaBu on the relative cell number and on the cell metabolism.
- Calculate the effect of NaBu as Resazurin / Hoechst responses.

## WEEK 2 EXPERIMENT

### Day 1: Seeding of DLD-1 and Caco-2 cells

- Use cells trypsinized in Exp 1 to seed.
- For Hoechst plate reader assay: Seed one 96 well plate with DLD-1 cells and one plate with Caco-2 cells: 10.000 cells per well. Seed in a volume of 200  $\mu$ L.

|            | 1                        | 2                    | 3                   | 4                   | 5                     | 6                              | 7                              | 8                              | 9                     | 10                             | 11                             | 12                       |
|------------|--------------------------|----------------------|---------------------|---------------------|-----------------------|--------------------------------|--------------------------------|--------------------------------|-----------------------|--------------------------------|--------------------------------|--------------------------|
| A          | No cells – media<br>only | Control:<br>PBS+DMSO | 1 mM NaBu +<br>DMSO | 5 mM NaBu +<br>DMSO | PBS + 10 $\mu$ M Resv | 1 mM NaBu + 10<br>$\mu$ M Resv | 5 mM NaBu + 10<br>$\mu$ M Resv | 5 mM NaBu + 20<br>$\mu$ M Resv | PBS + 40 $\mu$ M Resv | 1 mM NaBu + 40<br>$\mu$ M Resv | 5 mM NaBu + 40<br>$\mu$ M Resv | No cells – media<br>only |
| B          |                          |                      |                     |                     |                       |                                |                                |                                |                       |                                |                                |                          |
| C          |                          |                      |                     |                     |                       |                                |                                |                                |                       |                                |                                |                          |
| D          |                          |                      |                     |                     |                       |                                |                                |                                |                       |                                |                                |                          |
| E          |                          |                      |                     |                     |                       |                                |                                |                                |                       |                                |                                |                          |
| F          |                          |                      |                     |                     |                       |                                |                                |                                |                       |                                |                                |                          |
| G          |                          |                      |                     |                     |                       |                                |                                |                                |                       |                                |                                |                          |
| H          |                          |                      |                     |                     |                       |                                |                                |                                |                       |                                |                                |                          |
| Figure 2.2 |                          |                      |                     |                     |                       |                                |                                |                                |                       |                                |                                |                          |

### Day 2: Treatment with NaBu and resveratrol

- Use the concentrations of NaBu and Resv as indicated in the diagram below.  
Add a volume of 200  $\mu$ L for treatments (see Figure 2.2).

### Day 4 (Thursday):

- Perform the Hoechst 33342 and Alamar Blue assays as described for week 1.

## For the report – Exercise 2

- Include your laboratory notebook (scans of the relevant pages)

In relation to the cell cycle analysis

- Include all the dot plots, histograms and cell cycle distributions that you base your calculations on in your report. Prepare graphs for both WEEK 1 and WEEK 2 setups.
- You may calculate the cell cycle distribution using “flowing software” or “Flo-reada.io” (online)
- Describe the results of the cell cycle analysis: Is the cell cycle distribution different in the two cell lines and what was the effect of NaBu treatment? To answer this, you need to compare your results with the other groups’ results.

- How does the cell cycle analysis fit with the growth curve data from Report 1?
- What advantages and drawbacks are there for performing cell cycle analysis as a measurement of cell proliferation, compared with other methods, such as BrdU incorporation?

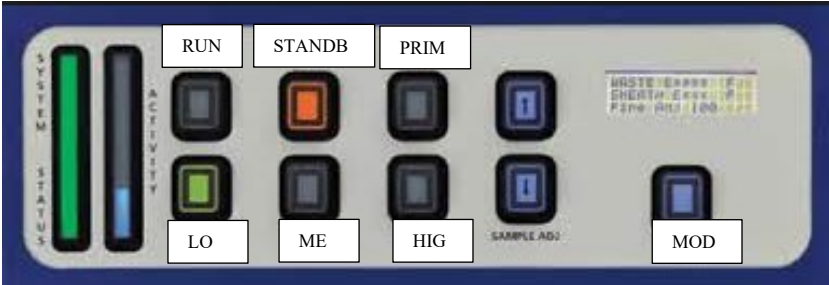
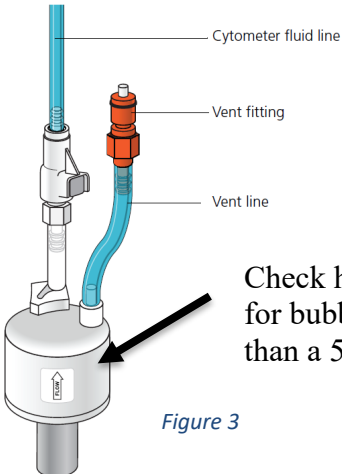
In relation to the 96-plate assays:

- Calculate and make a bar graph of mean readings with standard errors of Hoechst 33342 for each treatment and cell line for WEEK 1 and WEEK 2.
- Calculate and make a bar graph of mean readings with standard errors of /resazurin / Hoechst 33342 for each treatment and cell line for WEEK 1 and WEEK 2.
- Compare the cell counting results you obtained using hemocytometer and the Coulter counter (Report 1) with the results from the Hoechst-assay.
- What were the effects of NaBu and Resv assayed by Hoechst 33342 and did they depend on cell line? Compare results among groups.

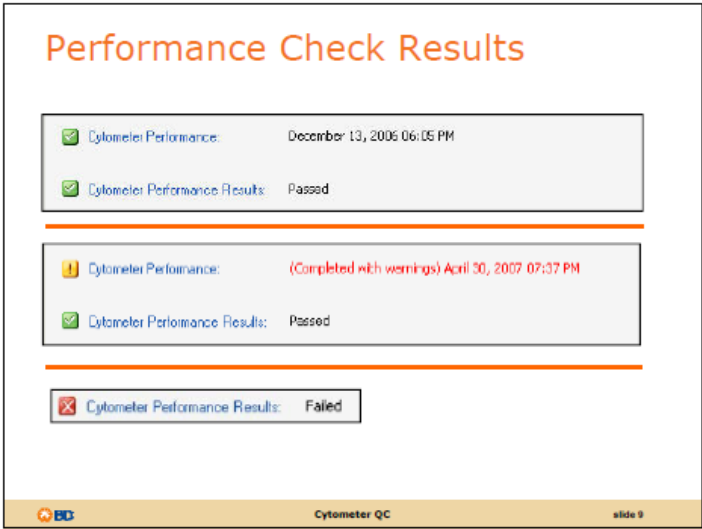
## Appendix 2.1 – Flow cytometer manual

### Protocol for simple setup Software (BDFACS DIVA) on Celesta FACS

| Startup                                | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Check Sheath tank and refill if needed | <ol style="list-style-type: none"> <li>1. Open the sheath tank placed on the floor. Do it by unscrewing the black part and gently tilt the lid. Refill with sheath fluid from the 20L container placed in the room. Fill up to the welded edge. Never fill it up to the lid.</li> <li>2. NB: <b>If you are running bacteria samples</b>, you need to change the sheath fluid from ordinary sheath fluid to water; Empty the contents by pouring out to a <u>clean and autoclaved</u> 10L Bluecap flask. Rinse the tank by adding autoclaved milli-Q-water/xx, pour it out, and fill the tank with autoclaved milli-Q-water/xx.</li> </ol> <div data-bbox="598 795 1401 1406" data-label="Image"> <p>Figure 1 shows a stainless steel sheath tank used for flow cytometry. It has a top lid with a black handle. A blue tube labeled 'Sheath fluid' is connected to the top. A 'Sheath fluid censor' is also connected. A 'Preussere valve' and a 'Preussere tube' are visible on the right side. A note 'For opening the tank' points to the top lid area.</p> </div> <p style="text-align: center;"><i>Figur 1</i></p> |
| Check waste tank and empty if needed   | <ol style="list-style-type: none"> <li>3. Empty the waste tank to a 5L Bluecap flask containing 500 mL deconex.</li> <li>4. This you may though out in the sink the day after your sample-run.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Turn on PC                             | 5. It is not necessary to turn on the FACS to do this                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Turn on the Celesta                    | 6. Big green bottom on the right side                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Lock into Windows as administrator     | 7. BDIS#1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Open BDFACSdiva software               | 8. Icon is on the "skrivebord". There is no password for this.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

|                                                                                                                                      |                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Press RUN on the Celesta</p>                                                                                                      | <p>9. To check if the machine creates pressure. Look at the Celeata left side; the System status have to show green – if red; tight the lid on the tank and see if it helps.</p>  <p><i>Figure 2</i></p>                              |
| <p>Check the sheath filter for air bubbles</p>                                                                                       |  <p><i>Figure 3</i></p> <p>10. If bubbles are visible, gently tap the filter to force them to the top, and then purge the air by pressing the small button at the end of the vent fitting until liquid flows out from the filter.</p> |
| <p>Check the sheath line for air bubbles. If necessary, open the roller clamp at the fluidics interconnect to bleed off any air.</p> | <p>11. Sheat line = the green/blue line ?? and the roller clamp is placed on the back of the mashine.</p> <p>12. At this time have a small tube ready and colect about ½ mL. of PBS from the tubing. You need this for later.</p>                                                                                       |
| <p>Prime the tubes with sheath fluid</p>                                                                                             | <p>13. Remove the DI water tube from the SIT.</p> <p>14. Press the PRIME button on the cytometer (FIG 2). Wait until the priming is complete and press PRIME <u>a second time</u>.</p>                                                                                                                                  |



|                                                                  |                                                                                                                                                                                                                                              |
|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                  |  <p style="text-align: right;">Aspirator arm</p> <p style="text-align: center;"><i>Figure 4</i></p>                                                        |
| Load a tube of DI water and press RUN, Set fluidics mode to HIGH | 15. Let the tube run for 10 minutes.                                                                                                                                                                                                         |
| Press standby                                                    | 16. (Always leave the SIT in a tube with clean milli-Q-water)                                                                                                                                                                                |
| <b>Run CST</b>                                                   | NB. Wait approximately 15-20 minutes after turn on (point 4)                                                                                                                                                                                 |
| Open CST Module                                                  | 17. ( <i>Cytometer</i> < <i>CST</i> )                                                                                                                                                                                                        |
| Prepare CST beads                                                | 18. 1 drop in 350uL PBS (from the small tube point 9)                                                                                                                                                                                        |
| Check Lot No on CST bead vial                                    | 19. Also verify that Lot No corresponds with Lot No in CST                                                                                                                                                                                   |
| Press Run on the Celesta                                         | 20. FIG 2                                                                                                                                                                                                                                    |
| Press Run in the CST Software                                    | 21. Event rate should start about 200 and then fluctuate while the CST program goes through the different test modules.                                                                                                                      |
| View report                                                      |  <p style="text-align: center;"><i>Figure 5</i></p> <p>22. Check DPMTV: 0 is the best</p> <p>23. Check Bright Bead rCV: check with last time reading</p> |

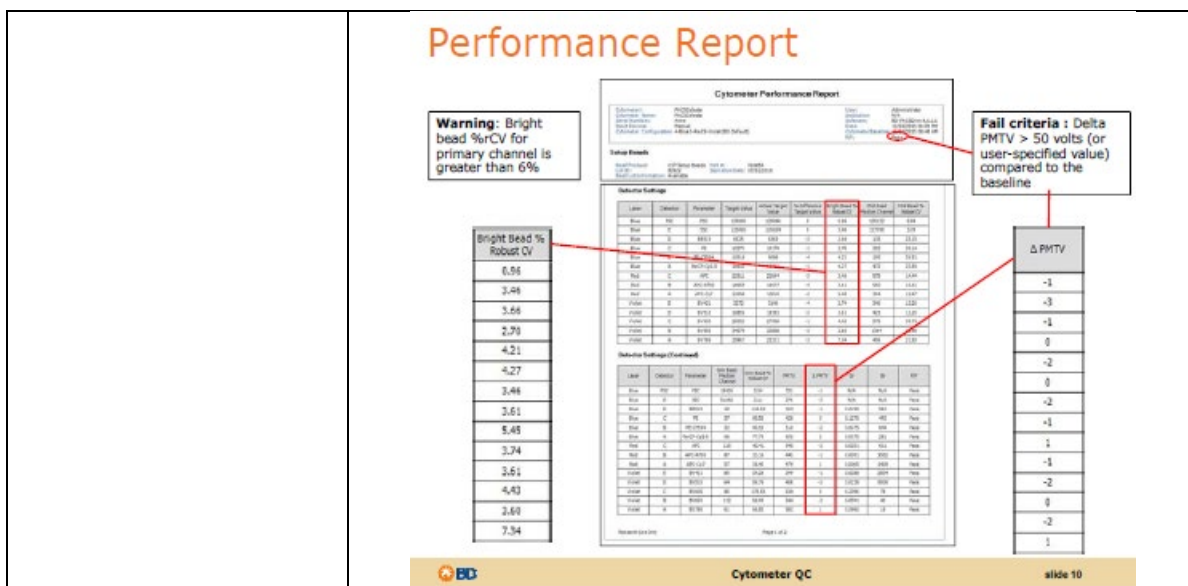


Figure 6

#### Setup software for your run

What to know before you can setup the software

24. Be sure you have all the information you need about the fluorophore (the excitation nm and the emission nm), how many cells you want to measure.

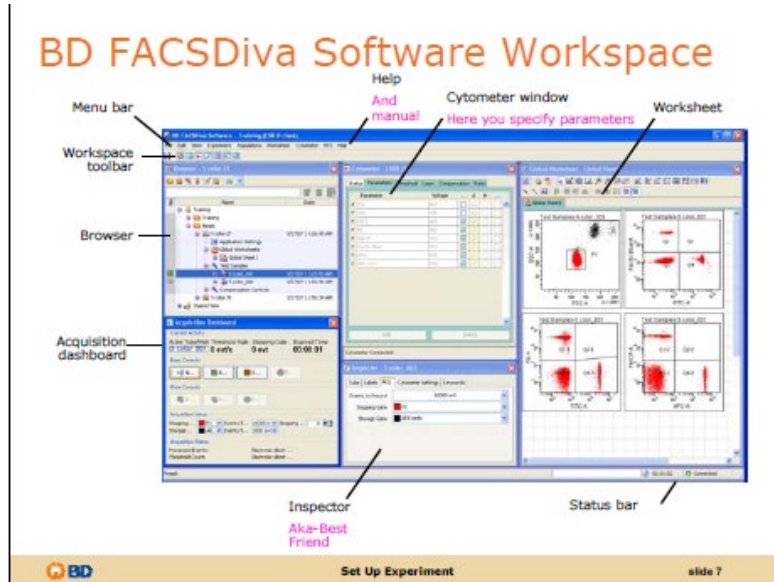
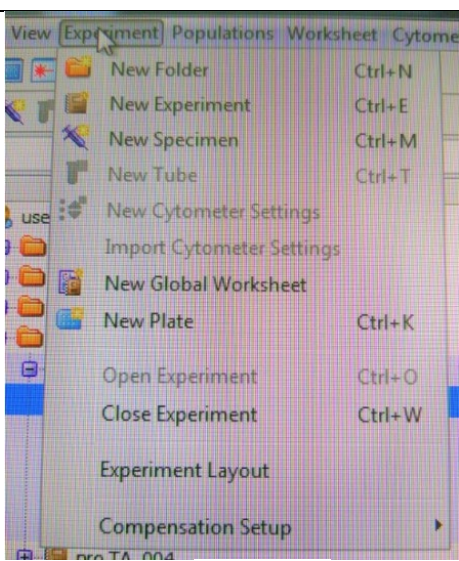


Figure 7

Choose NEW EXPERIMENT

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <ul style="list-style-type: none"> <li>• Blank experiment: only for Areal</li> <li>• <b><u>Blank experiment with sample tube: Areal, height and wide</u></b></li> <li>• Blank experiment with sample tube_003: PE-CF595-A/PE-A</li> <li>• Pro TA_005: For RIF-run-out</li> </ul>  <p style="text-align: right;"><i>Figure 8</i></p> <p>25. These will be your template for your run<br/><i>remember to save your experiment with a new name</i></p> |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Open VIEW</p> <p>Make sure Toolbar, Status bar, Browser, Cytometer, Inspector, Acquisition and Worksheet is present.</p> | <p>26. <b>Toolbar:</b> Icons for the most common used functions</p> <p>27. <b>Statusbar:</b> you can see if you have lost connection to the Celesta.</p> <p>28. <b>Browser:</b> Like the tree of folders and experiments</p> <p>29. <b>Cytometer:</b> XXX</p> <p>30. <b>Inspector:</b> Write the exp. name (fil name) and choose if you want 4 or 5 decades<br/>If the cytometer window is active the inspector shows that options you want active/deactivated; A=areal, H=high, W=wide.</p> <p>31. <b>Acquisition:</b> Is where you start the connection between the software and the celesta by pressing acquirer. You can decide how many cells in each sample you want to count ex. 10,000 and it also shows the electronic aborts</p> <p>32. <b>Worksheet (Global worksheet):</b><br/>Choose the different DOT BLOT, CONTOUR and HISTOGRAMS.</p> |
|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Choose the detector you want to use. To do so you have to know on what wavelength your specific fluorochrome will emit light.

**Check out the table posted beside the of the workbench**

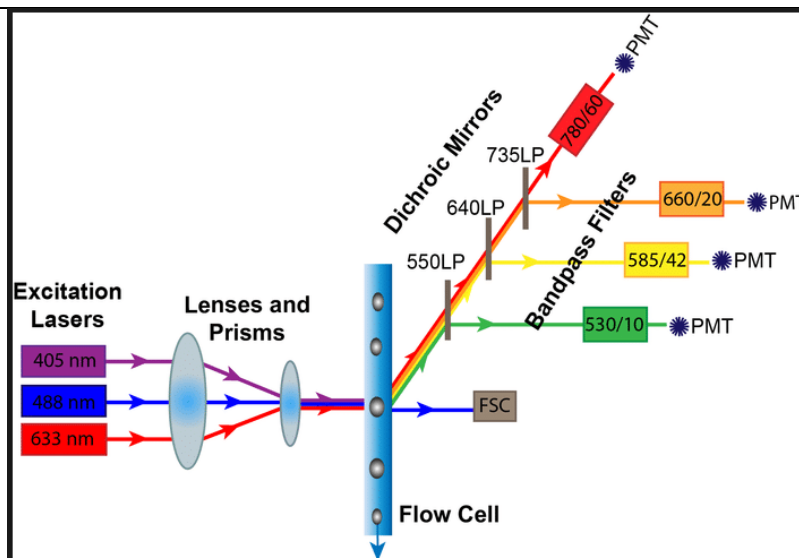


Figure 9 Example of how the light will operate inside the FACS

This is an example of setup; all the different graphs needed and the operator starts to optimize what filters to use, if the axes should be log or not, PMT values up and down. Use the Cytometer menu for regulating the PMT values etc.

### Worksheet

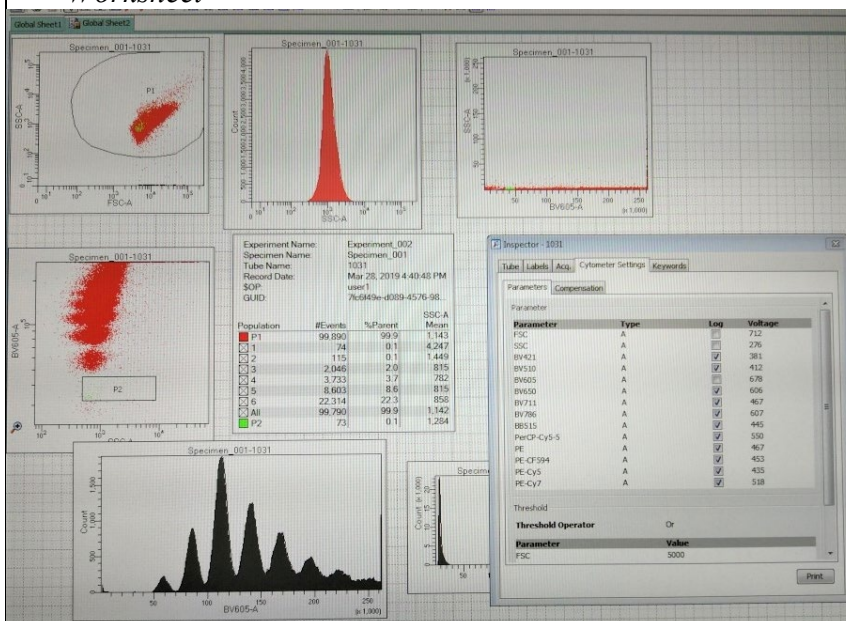
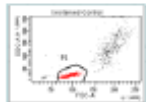
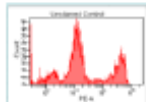
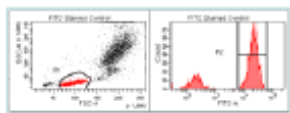
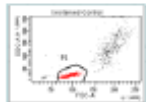
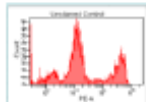
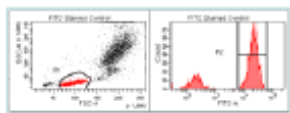
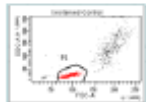
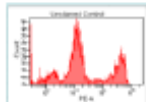
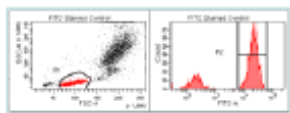


Figure 10

| <p><b>How to do that:</b></p>                                                                                                                                                                                                                    | <p><b>Set Up Experiment:</b><br/>Adjusting cytometer settings for population of interest</p> <table border="1"> <thead> <tr> <th>Steps – Start with CS&amp;T Settings</th><th>Example</th></tr> </thead> <tbody> <tr> <td> <b>1.</b> Adjust scatter settings.<br/> <ul style="list-style-type: none"> <li>- Adjust FSC and SSC voltages.</li> <li>- Adjust threshold.</li> <li>- Gate the population of interest.</li> </ul> </td><td></td></tr> <tr> <td> <b>2.</b> Verify PMT voltages for fluorescence parameters.<br/> <ul style="list-style-type: none"> <li>- Check negative populations.</li> <li>- Check positive populations so that bright positive populations have MFIs below 100,000.</li> </ul> </td><td></td></tr> <tr> <td> <b>3.</b> Save Application Settings (optional).<br/> <ul style="list-style-type: none"> <li>- Right-click Cytometer Settings &gt; Select Application Settings &gt; Save</li> </ul> </td><td></td></tr> <tr> <td> <b>4.</b> Calculate compensation.<br/> <ul style="list-style-type: none"> <li>- Record data for compensation controls.</li> <li>- Verify gates.</li> <li>- Select Experiment &gt; Compensation Setup &gt; Calculate Compensation.</li> </ul> </td><td></td></tr> </tbody> </table> <p> <b>Advanced Optimization and Application Settings</b> <span style="float: right;">slide 12</span></p> <p><i>Figur 11</i><br/>Note: Calculate compensation is only necessary if you are staining with more than one stain!</p> | Steps – Start with CS&T Settings | Example | <b>1.</b> Adjust scatter settings.<br><ul style="list-style-type: none"> <li>- Adjust FSC and SSC voltages.</li> <li>- Adjust threshold.</li> <li>- Gate the population of interest.</li> </ul> |  | <b>2.</b> Verify PMT voltages for fluorescence parameters.<br><ul style="list-style-type: none"> <li>- Check negative populations.</li> <li>- Check positive populations so that bright positive populations have MFIs below 100,000.</li> </ul> |  | <b>3.</b> Save Application Settings (optional).<br><ul style="list-style-type: none"> <li>- Right-click Cytometer Settings &gt; Select Application Settings &gt; Save</li> </ul> |  | <b>4.</b> Calculate compensation.<br><ul style="list-style-type: none"> <li>- Record data for compensation controls.</li> <li>- Verify gates.</li> <li>- Select Experiment &gt; Compensation Setup &gt; Calculate Compensation.</li> </ul> |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Steps – Start with CS&T Settings                                                                                                                                                                                                                 | Example                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                  |         |                                                                                                                                                                                                 |                                                                                     |                                                                                                                                                                                                                                                  |                                                                                     |                                                                                                                                                                                  |  |                                                                                                                                                                                                                                            |                                                                                     |
| <b>1.</b> Adjust scatter settings.<br><ul style="list-style-type: none"> <li>- Adjust FSC and SSC voltages.</li> <li>- Adjust threshold.</li> <li>- Gate the population of interest.</li> </ul>                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                  |         |                                                                                                                                                                                                 |                                                                                     |                                                                                                                                                                                                                                                  |                                                                                     |                                                                                                                                                                                  |  |                                                                                                                                                                                                                                            |                                                                                     |
| <b>2.</b> Verify PMT voltages for fluorescence parameters.<br><ul style="list-style-type: none"> <li>- Check negative populations.</li> <li>- Check positive populations so that bright positive populations have MFIs below 100,000.</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                  |         |                                                                                                                                                                                                 |                                                                                     |                                                                                                                                                                                                                                                  |                                                                                     |                                                                                                                                                                                  |  |                                                                                                                                                                                                                                            |                                                                                     |
| <b>3.</b> Save Application Settings (optional).<br><ul style="list-style-type: none"> <li>- Right-click Cytometer Settings &gt; Select Application Settings &gt; Save</li> </ul>                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                  |         |                                                                                                                                                                                                 |                                                                                     |                                                                                                                                                                                                                                                  |                                                                                     |                                                                                                                                                                                  |  |                                                                                                                                                                                                                                            |                                                                                     |
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| <p><b>Run your samples</b></p>                                                                                                                                                                                                                   | <p>33. Click on the tiny triangle on the left side of the tube icon to collect data in this file in the browser menu</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                  |         |                                                                                                                                                                                                 |                                                                                     |                                                                                                                                                                                                                                                  |                                                                                     |                                                                                                                                                                                  |  |                                                                                                                                                                                                                                            |                                                                                     |
|                                                                                                                                                                                                                                                  | <p>34. Wortex your sample</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                  |         |                                                                                                                                                                                                 |                                                                                     |                                                                                                                                                                                                                                                  |                                                                                     |                                                                                                                                                                                  |  |                                                                                                                                                                                                                                            |                                                                                     |
|                                                                                                                                                                                                                                                  | <p>35. Place the SIT in the sample tube by</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                  |         |                                                                                                                                                                                                 |                                                                                     |                                                                                                                                                                                                                                                  |                                                                                     |                                                                                                                                                                                  |  |                                                                                                                                                                                                                                            |                                                                                     |
|                                                                                                                                                                                                                                                  | <p>36. Press RUN on the celesta and press Acquirer in the Acquisition panel</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                  |         |                                                                                                                                                                                                 |                                                                                     |                                                                                                                                                                                                                                                  |                                                                                     |                                                                                                                                                                                  |  |                                                                                                                                                                                                                                            |                                                                                     |
|                                                                                                                                                                                                                                                  | <p>37. Wait until the event/sec is stable and then press record in the acquisition panel.<br/>When you do so you start collection data.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                  |         |                                                                                                                                                                                                 |                                                                                     |                                                                                                                                                                                                                                                  |                                                                                     |                                                                                                                                                                                  |  |                                                                                                                                                                                                                                            |                                                                                     |
| <p>Saving data and exporting data</p>                                                                                                                                                                                                            | <p>38. Samples are saved automatically; you see small diskettes beside the tube icon.<br/>If you want to export your files: Select File – Export and XXXXX</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                  |         |                                                                                                                                                                                                 |                                                                                     |                                                                                                                                                                                                                                                  |                                                                                     |                                                                                                                                                                                  |  |                                                                                                                                                                                                                                            |                                                                                     |
| <p>Free software's for analysing the FCS datafiles</p>                                                                                                                                                                                           | <p>Flowing software – download and install on your own computer:<br/> <a href="http://flowingsoftware.btk.fi/index.php?page=4">http://flowingsoftware.btk.fi/index.php?page=4</a><br/> <a href="http://flowingsoftware.btk.fi/tutorial/tutorial_video.wmv">http://flowingsoftware.btk.fi/tutorial/tutorial_video.wmv</a><br/> <br/> Floreada.io (online free ware):<br/> <a href="https://floreada.io/analysis">https://floreada.io/analysis</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                  |         |                                                                                                                                                                                                 |                                                                                     |                                                                                                                                                                                                                                                  |                                                                                     |                                                                                                                                                                                  |  |                                                                                                                                                                                                                                            |                                                                                     |

|                            |                                                                                                                                                                                                                                                                         |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Shutdown</b>            | Very important: IF you have used ethidium bromide, you must clean the instrument in a special way: Repeat no. 31 – 39 ones more.<br>Prepare 1 tube with ~ 4 mL for each: milli-Q-water, FACS Clean, FACS Rinse and an extra fresh tube filled with fresh milli-Q-water. |
| Clean the flow cell        | 39. Place a tube containing FACS Clean in the SIT                                                                                                                                                                                                                       |
|                            | 40. Press PRIME.                                                                                                                                                                                                                                                        |
|                            | 41. When you see air bubbles coming from the SIT, immediately press run                                                                                                                                                                                                 |
|                            | 42. Repeat the procedure (b+c) at least 3 times.                                                                                                                                                                                                                        |
|                            | 43. Place a tube containing DI water (Millipore) in the SIT and repeat the procedure (b+c) at least 3 times more.                                                                                                                                                       |
| Perform the fluidics clean | 44. Place a tube with DI water in the SIT and move aspirator arm to side for 1 minute, move arm back in position and run at high flow rate for 3-5 minutes.<br>45. Repeat with FACS Clean<br>46. Repeat with FACS Rinse<br>47. Repeat with DI water                     |
|                            | 48. Place a tube of DI water on the SIT.                                                                                                                                                                                                                                |
|                            | 49. Turn off the Celesta.                                                                                                                                                                                                                                               |
|                            | 50. Close Diva and shut down the computer.                                                                                                                                                                                                                              |



## Exercise 3 – Transfection and luciferase assay

### **Description of CRISPR - luciferase system**

Genetic reporter systems are widely used to study eukaryotic gene expression and cellular physiology. Applications include the study of receptor activity, transcription factors, intracellular signaling, mRNA processing and protein folding.

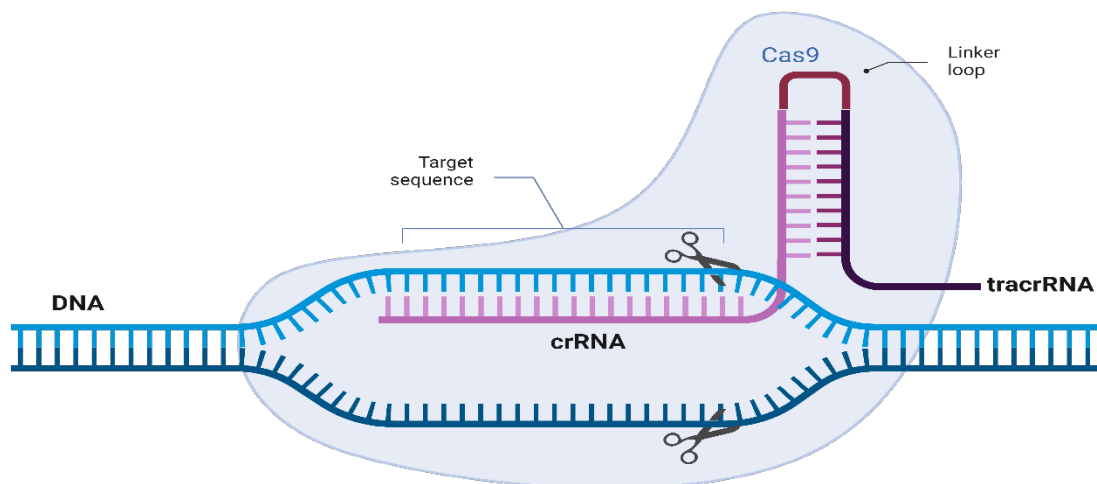
A reporter gene assay indirectly measures the activity of a chosen promoter. Thus, a piece of promoter is cloned in front of a reporter gene, whose product is easy to measure in a quantitative assay. The activity of the reporter gene (which most often is luciferase but may also be  $\beta$ -galactosidase, secreted alkaline phosphatase, chloramphenicol acetyl transferase or other) is proportional to the activity of the promoter element cloned in front of it.

The amount of luciferase activity produced depends on promoter activity but also on factors such as how many cells that took up the reporter plasmid and how many copies of the reporter plasmid that each cell has after the transfection. This technical variation can be very large and will affect the interpretation of the results if not accounted for. Therefore, we co-transfect the cells with a plasmid that contains Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR).

CRISPR is an effective tool for genetic engineering, enabling precise and efficient editing of DNA. It was originally discovered as part of the immune system in bacteria, enabling the bacteria to capture snippets of viral DNA and inserts them into its genome at the CRISPR loci, thereby serving as genetic memory protecting the bacterium when reencountering the same virus.

The CRISPR system function by a single synthetic RNA called single-guide RNA (sgRNA) which is combined of CRISPR RNA (crRNA) and trans-activating crRNA (tracrRNA). The crRNA is the RNA sequence complementary to the target DNA sequence,

and the tracrRNA is an RNA sequence that binds to the crRNA, guiding specialized proteins (Cas9) to the targeted DNA location. The sgRNA can be designed to match any specific DNA sequence. The sgRNA forms a complex with the Cas9 protein guiding it to the specific location, where it makes a double-strand break in the DNA at the target site. These breaks can be repaired by either non-homologous end joining or homology-directed repair, depending on the desired experimental outcome. The non-homologous end joining will result in unspecific insertions or deletions, whereas homology-directed repair uses a DNA template to ensure a specific and precise mutation or insertion.



In this experiment, you will perform a transfection with and without CRISPR to assess the efficiency of CRISPR-mediated gene editing by using Lucia luciferase as reporter gene. Thus, comparing the luciferase activity in cells transfected with CRISPR versus cells without CRISPR. You will co- transfect the Caco-2 cells (week 1) and DLD-1 cells (week 2) with a plasmid (AAVS1-T2-CRISPR in pX330) containing the CRISPR system targeted the AAVS1 (a safe harbour locus) and a plasmid with Lucia luciferase designed for the AAVS1 (pSelect-Zeo-Lucia-AAVS1). This should result in a stable transfection



by specific insertion of the Lucia luciferase reporter construct in the AAVS1 locus by homology directed repair. We use a synthetically designed Lucia gene obtained from marine copepods, which produces a secreted form of luciferase. This luciferase, an oxidative and bioluminescent enzyme, is specifically designed for stable and sustained expression in mammalian cells. The Lucia gene is controlled by the EF-1 $\alpha$ /HTLV promoter, which is designed to provide this robust and continuing expression of transgenes, surpassing the activity of several other promoters, such as the CMV. The strong and stable expression makes it especially suitable for *in vitro* and *in vivo* transfection experiments. The Lucia luciferase provides a 1000-fold higher bioluminescent activity than the commonly used luciferases from Firefly and Renilla, making it easy to detect in the cell culture growth media. It uses coelenterazine as substrate, emitting visible blue light between 465 and 493 nm, and does not require ATP or Mg<sup>2+</sup> for light production.

### **Measuring luciferase activity**

The measurement of luciferase will be done with the QUANTI-Luc assay reagent and transferring 20  $\mu$ L growth media from each well to a 96-well microplate and measure luciferase activity. The assay can be detected on any machine capable of quickly measuring light emission. We use the GloMax Luminometer, which has two injectors, but for this measurement we will only use injector one set to 50  $\mu$ L injections and end-point measurements with 4 s delay, and 0.5 s reading time.

Transfection of plasmid DNA into mammalian cells can be carried out in a number of ways. Many transfection reagents are commercially available. For most experiment, we use the linear Polyethyleneimine (PEI) with the molecular weight of 25 kDa. It functions

well in our preferred cell lines and is cheaper than other reagents sold as transfection reagent.

### **Materials**

1. Caco-2 cells (week 1)
2. DLD-1 cells (week 2)
3. 6 well Nunclon Delta treated cell culture plates
4. 85 mM sterile Sodium citrate (SC), sterile filtered.
5. PBS w/o Ca and Mg (PBSw/o)
6. 150 mM NaCl, sterile filtered.
7. Growth medium, complete: DMEM supplemented with 10% Fetal Bovine Serum (FBS), and 100 U/ mL Penicillin-Streptomycin (P/S)
8. 2  $\mu$ M Polyethyleneimine (PEI) (Alfa Aesar)
9. Plasmids for promoter reporter assays:
  - a. AAVS1-T2-CRISPR in pX330
  - b. pSelect-Zeo-Lucia-AAVS1
  - c. pUC18: Plasmid without eukaryotic gene regulatory sequence domains.
10. Phosphate Buffered Saline with  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$  (PBSw)
11. Quanti-luc
12. GloMax-96 Microplate Luminometer with dual injectors (Promega).
13. White 96-well plates with flat bottom (Costar).
14. 70% ethanol.

## Experiment 3 – Transfection and luciferase assay

### WEEK 1 EXPERIMENT: TRANSFECTION AND LUCIFERASE ASSAY SETUP

#### Day 1: Seeding of Caco-2 cells

Use cells trypsinized in Exp 1 to seed all 6 wells in 6 well plate with each 200.000 cells/well in a volume of 4000  $\mu$ L/well.

- Mock transfection: Two wells for pUC18 only.
- Control: Two wells for pSelect-Zeo-Lucia-AAVS1
- Positive: Two wells for pSelect-Zeo-Lucia-AAVS1 and AAVS1-T2-CRISPR in pX330

#### Day 2: Transfection with linear polyethyleneimine (PEI) and plasmids

##### DNA-NaCl mix:

For each well in every transfection condition a total of 1.4  $\mu$ g DNA is needed in 125  $\mu$ l 150 mM NaCl. The plasmid DNA has been pre-diluted in 150 mM NaCl to:

pSelect-Zeo-Lucia-AAVS1: 35,6 ng/ $\mu$ l  
 AAVS1-T2-CRISPR in pX330: 46,5 ng/ $\mu$ l  
 pUC18 + : 105 ng/ $\mu$ l

Use the diluted DNA for the following DNA mixes and add pUC18 to a total of 1.4  $\mu$ g DNA in each mix and fill up with 150 mM NaCl to a final volume of 125  $\mu$ l pr well.

| DNA in $\mu$ g      | Pselect-Zeo-Lucia | AAVS1 T2 CRISPR | pUC18 | Total |
|---------------------|-------------------|-----------------|-------|-------|
| Control Lucia alene | 1.12              | 0.00            | 0.28  | 1.4   |
| Lucia + Crispr      | 1.05              | 0.35            | 0.00  | 1.4   |
| Mock                | 0.00              | 0.00            | 1.4   | 1.4   |

| Transfer in $\mu$ l | Pselect-Zeo-Lucia | AAVS1 T2 CRISPR | pUC18 | Total | Adjust to 250 $\mu$ l with NaCl (150 mM) |
|---------------------|-------------------|-----------------|-------|-------|------------------------------------------|
| Control             |                   |                 |       |       |                                          |
| CRISPR              |                   |                 |       |       |                                          |
| Mock                |                   |                 |       |       |                                          |

Mix DNA and PEI solutions to form transfection complexes:

1. One hour prior to transfection mix the 250  $\mu$ l DNA solution with 250  $\mu$ l 2  $\mu$ M PEI for each transfection condition. Vortex briefly and let the reaction stand to form precipitates at room temperature.
2. Remove media of cells seeded in the 6 well plate and add 4 mL fresh supplemented DMEM media to each well.
3. Add in small droplets, 245  $\mu$ l of DNA / PEI mixture to the cells in each well. Avoid touching plastic surfaces. To get even amounts of DNA: PEI precipitates to the wells remember to mix it by pipetting up and down before taking out.
4. Incubate in CO<sub>2</sub> incubator.

**Day 3: Media change**

1. Change medium.

**Day 4: Transfer media and perform luciferase assay**

1. All steps could be performed outside the LAF bench.
2. Remove cells from the incubator.
3. Look at the cells in the microscope.

**Luciferase assay**

1. Transfer 20  $\mu$ L of growth media from each well into individual wells of a white flat bottom 96 well plate (also label these). Groups may share the same 96 well luminometer plate – just mark where your samples are located. Plate should be room temperature before initiating measurement.
2. Before measuring, prepare the Quanti-luc. The instructor will do this and show you how according to the manufacture's instructions.
3. Prepare the luminometer. The luminometer should now be rinsed and initiated (which the instructor will do but will be happy to show).
4. Perform the luciferase assay with GloMax96 luminometer using the quanti-luc settings.

#### **Day 5: Seeding of the transfected Caco-2 cells for luciferase proliferation assay**

1. Trypsinize and count the cells as described on **page 19-22**. Record 1) counts, cells/mL, 3) number of cells in total
2. Seed transfected Caco-2 5.000 cell/mL, 1.0 mL/well in 24-well plate in column 1, 5 mL is enough for all of the column.

### **WEEK 2 EXPERIMENT: TRANSFECTION AND LUCIFERASE ASSAY SETUP**

#### **Day 1: Seeding of DLD-1 cells**

Use cells trypsinized in Exp 1 to seed. Seed 6 wells in 6 well plate with each 200.000 cells/well in a volume of 4000  $\mu$ L/well.

- Mock transfection: Two wells for pUC18 only.
- Control: Two wells for pSelect-Zeo-Lucia-AAVS1
- Positive: Two wells for pSelect-Zeo-Lucia-AAVS1 and AAVS1-T2-CRISPR in pX330

Measure luciferase activity on transfected caco-2 cells as described in week 1

#### **Day 2: Transfection with linear polyethyleneimine (PEI) and plasmids**

DNA-NaCl mix:

For each well in every transfection condition a total of 1.4 µg DNA is needed in 125 µl 150 mM NaCl. The plasmid DNA has been pre-diluted in 150 mM NaCl to:

pSelect-Zeo-Lucia-AAVS1: 35,6 ng/µl  
 AAVS1-T2-CRISPR in pX330: 46,5 ng/µl  
 pUC18 + : 105 ng/µl

Use the diluted DNA for the following DNA mixes and add pUC18 to a total of 1.4 µg DNA in each mix and fill up with 150 mM NaCl to a final volume of 125 µl pr well.

| DNA in µg           | Pselect-Zeo-Lucia | AAVS1 T2 CRISPR | pUC18 | Total |
|---------------------|-------------------|-----------------|-------|-------|
| Control Lucia alene | 1.12              | 0.00            | 0.28  | 1.4   |
| Lucia + Crispr      | 1.05              | 0.35            | 0.00  | 1.4   |
| Mock                | 0.00              | 0.00            | 1.4   | 1.4   |

| Transfer in µl | Pselect-Zeo-Lucia | AAVS1 T2 CRISPR | pUC18 | Total | Adjust to 250 µl with NaCl (150 mM) |
|----------------|-------------------|-----------------|-------|-------|-------------------------------------|
| Control        |                   |                 |       |       |                                     |
| CRISPR         |                   |                 |       |       |                                     |
| Mock           |                   |                 |       |       |                                     |

Mix DNA and PEI solutions to form transfection complexes:

- One hour prior to transfection mix the 250 µl DNA solution with 250 µl 2 µM PEI for each transfection condition. Vortex briefly and let the reaction stand to form precipitates at room temperature.

6. Remove media of cells seeded in the 6 well plate and add 4 mL fresh supplemented DMEM media to each well.
7. Add in small droplets, 245  $\mu$ l of DNA / PEI mixture to the cells in each well.  
Avoid touching plastic surfaces. To get even amounts of DNA: PEI precipitates to the wells remember to mix it by pipetting up and down before taking out.
8. Incubate in CO<sub>2</sub> incubator.
9. Measure luciferase activity on transfected Caco-2 cells as described in week 1

**Day 3: Media change**

1. Change medium.
2. Measure luciferase activity on transfected Caco-2 cells as described in week 1

**Day 4: Transfer media and perform luciferase assay**

Measure luciferase activity as described in week 1 on both DLD-1 cells and the transfected Caco- 2 cells

## For the report - Exercise 3

- Include your laboratory notebook (scans), raw data in table format and annotated images.
- Describe visual effects (if any) on the cells following transfection (did they die, become non-adherent or change shape?)
- Plot the luciferase activity (RLU – relative light units) of each sample.
- Describe your findings. What do you expect in your different transfections? Do you see a completely equal activity of luciferase activity in all your samples? Is that expected or is there an unexpected pattern?
- Compare the luciferase measurements from both cell lines and plot as box plot with error bars.
- Plot the luciferase measurements from the luciferase proliferation assay inclusive standard deviations and describe your findings.



## Appendix 3.1 Luciferase measurement on Glomax luminometer

### Luciferase measurement on Glomax luminometer - Protocol

3. Transfer 20 $\mu$ L of growth media from each sample to a 96 well plate.

#### Measuring luciferase:

Before measuring, make sure that the Quanti-luc have been prepared and that you have the appropriate white flat bottom 96-well plates for the luminometer.

1. Turn on the Glomax machine and open the program on the computer. Following the cleaning process under Tools and the injector settings. Remember to flush 3 times at each cleaning step.
2. Press 'open saved protocol' and choose 'Quanti-luc (saved in 'Glomax protocols')'.
3. Place Quanti-luc tube in the machine and connect injector tubes. Prime injector 1.
4. Put the 96 well plate in the machine. If you do not have a complete 96 well plate, choose which wells you are measuring in the options menu. While in the options menu, check that injection volume for injector 1 is set at 50  $\mu$ L.
5. Press start. When all the wells have been measured, save your data by pressing 'save a copy as' in the menu tab.
6. Reverse purge injector 1
7. 'Flush' and follow the listed steps. Remember to flush 3 times at each cleaning step.

## Exercise 4a – Fluorescent imaging and staining

### Introduction

A nice microscope image always makes applause! Further, immunocytochemistry is an important way to study the specific structures of the cells and a powerful tool together with relevant molecular biology techniques.

Immunocytochemistry uses labeled antibodies that target antigenic epitopes on the cell membrane or intracellularly. There are many methods to perform immunological stains on cells (and tissue), including the direct method of staining. The direct labeling involves the use of a fluorescent or colored dye bonded directly to the antibody, which binds to the antigen. Beside this, there are many indirect methods. In one such method, the primary antibody binds to the antigen, and a secondary antibody coupled with the dye binds to the primary antibody. An optimal and specific signal is dependent on the specificity of the antibodies used (both primary and secondary) and the affinity of the antibodies used. In fact, we are not using an antibody in these experiments, but phalloidin (a toxic compound) from the death cap mushroom (*Amanita phalloides*), which specifically binds to polymerized actin. Therefore, it is possible to detect polymerized actin (Week 2), Other ways to visualize cell structures is to transfect cells with fluorescent proteins targeted to specific cellular structures or organelles. Several fluorescent dyes may stain specific organelles, as we are using MitoTrackerGreen to live-stain mitochondria and Hoechst 33342 or DAPI to stain nuclei as they binds to DNA.

**Fixed cells:** Cells may be fixed using different methods, where fixatives as formaldehyde and glutaraldehyde are the most used. They have their advantages and disadvantages. Formaldehyde quickly penetrates the cell membrane and reacts with nitrogen groups in the cellular proteins. Glutaraldehyde has a higher degree of fixation capacity compared with formaldehyde. Further, a secondary fixation may be relevant where, e.g., cells are first fixated with glutaraldehyde, and then the osmium tetroxide will penetrate the cell,

where it reacts with lipids, stabilizing the cell membrane as well as the membranes of organelles. To get antibodies or dyes into cells, cell membranes have to be permeabilized, and for this Triton X-100 is often used.

Several fluorescent dyes are available to differently label antibodies. Thus, you can stain three (or more) different proteins for localization at the same time. You may also combine the immunocytochemistry with a specific stain of an organelle (e.g. mitochondria or nucleus).

In week 2, we stain cells for cytoskeletal structures after fixation of the cells.

**Live cell imaging** (without fixation) can be done using fluorescent dyes localizing to different intracellular compartments or responding to different cellular signaling events, such as changes in Calcium ion concentrations (measured for example using Fura2), or ATP generation (measured using PercevalHR (Addgene #49082) fluorescent protein signals). In Week 1, we will quantify nuclear size in response to NaBu and Resv treatment using **live cell imaging** with the cell permeable dye Hoechst 33342.

In the present experiment you expose human colon cells, DLD-1 or Caco-2 to Resv (20  $\mu$ M) or NaBu (1 mM) for 72hrs and identify the effects on cell size, nucleus size and cytoskeletal structure by microscopy. In the cited paper [1], it was shown that for a short time exposure (10 min) Resv also modulates the number of filopodia and lamellopodia.

## Experiment

### Materials

1. Glass cover slips. Each group needs at least 8 coverslips. Coat a few extras.
2. 2M NaOH (aq)
3. Poly-D-lysine (Sigma, P-6407).
4. Glass petri dishes
5. 70% EtOH
6. 96% EtOH
7. 35 mm Ibidi dishes (170  $\mu$ m glass bottoms, so be careful)

8. DLD-1 and Caco-2 cells: Team 1+2 will use DLD1 cells and team 3 will use Caco-2 cells.
9. 12 well cell culture plates
10. 85 mM sterile Sodium citrate (SC)
11. PBS w/o  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$
12. Object glass
13. Growth medium, complete for both DLD-1 and Caco-2 cells
14. Resveratrol (600 mM)
15. Sodium butyrate (1 M)
16. DMSO
17. Scalpels, non-sterile, preferably with round edges
18. 800  $\mu\text{M}$  Calcein Red-Orange, AM
19. 400  $\mu\text{M}$  MitoTracker Green
20. 10 mg/ mL Hoechst 33342
21. PBSw + 10 mM glucose
22. 10% formalin in PBS for fixation
23. BSA
24. Rhodamin-Phalloidin (Molecular Probes) (TRITC conjugated: Excitation/emission: 540 nm /565 nm
25. ProLong™ Glass Antifade Mountant with NucBlue™ Stain (Hoechst 33342) (Thermo Fisher Scientific P36983)
26. Nail polish
27. Inverted fluorescence microscope (Leica IM DRB / EVOS M5000) or confocal microscope (Olympus FV-1200)

Prior to experiment start, poly-lysine coated cover slips must be prepared. No coating is necessary for the 35 mm Ibidi dishes (live-imaging).

## **WEEK 1 EXPERIMENT: LIVE CELL FLUORESCENT IMAGING OF NUCLEI**

**Day 1: Seeding of DLD-1 (Team 1+2) or Caco-2 (Team 3+4) cells**

- Use four Ibidi dishes / team
- Use cells trypsinized in Exp 1 to seed. Seed 50.000 cells per dish in a total of 2.5 mL medium.

**Day 2: Treatment with resveratrol and NaBu**

- Prepare growth medium containing 1 mM NaBu and/or 20  $\mu$ M Resv. This will create 4 conditions:
  - Control (add PBSw and DMSO in the same amounts as in treatments)
  - 1 mM NaBu (include DMSO as control)
  - 20  $\mu$ M Resv (include PBSw as control)
  - 1 mM NaBu and 20  $\mu$ M Resv

We expect to be able to observe an effect of Resv after 48-72 hrs, and will therefore image the cells after 72 hrs, but will briefly observe at them at Day 1, 2 and Day 4. Image cells briefly, because extensive time away from the cell incubator impairs cell viability. Acquire a few images for documentation as long as the cells are not out of the incubator for more than 10 min. Using the confocal microscope, an incubator is part of the microscope why we may have living cells there for a longer time.

**Day 5: Stain and acquire images of cells using fluorescence microscope**

1. Microscope in room 28A.2-37: Leica DM IRB / EVOS M5000 or confocal Olympus microscope (28AB.0-07)
2. Look at cells in phase contrast microscopes to gain information on viability.
3. If cells are viable and adhere well to bottom, then stain cytosol with Calcein Red-Orange, AM and mitochondria with MitoTracker Green
  - a. Remove media – wash 2x with fresh media
  - b. Add 2 mL medium / dish containing
    - i. Add 2  $\mu$ L 800  $\mu$ M Calcein Red-Orange, AM / 2 mL media.
    - ii. Add 2  $\mu$ L 400  $\mu$ M Mito Tracker Green / 2 mL media.
4. Incubate for 60 min

5. After 45 min with Calcein and MitoTracker, counter stain with Hoechst 33342
  - a. Use 2 drops / 2 mL (1  $\mu$ L 10 mg/mL Hoechst 33342/ 2 mL media)
6. Incubate the cells for additional 15 min.
7. Remove media and add PBSw with 10 mM glucose instead.
8. Then check cells under microscope to assess good staining.
9. Magnification: Use the 20x and 40x objectives.
10. Fluorescence filters:
  - a. DAPI filter (white light, blue filter) for Hoechst 33342 (Ex 360/40 nm / EM 460/40 nm)
  - b. Rhodamine filter for Calcein RedOrange (Ex 577/10 nm; Em: 590/20 nm)
  - c. GFP filter for Mito Tracker Green (Ex 485/20 nm / EM 540/25 nm)
11. Fading. Be aware that especially red colors fade quickly. For our purposes, Hoechst is quite stable. Nevertheless, expose the areas of interest as short time as possible to fluorescent light, before acquiring images.
12. Identify the cells with phase contrast. Locate cells that do not touch neighboring cells. Identify the relevant focal plane. Switch to the fluorescence and acquire image. For more extensive microscopy, it is also possible to acquire several pictures at the same position, but you might have to increase the exposure time. Be sure to acquire images of at least 20 nuclei of each condition.
13. Label images with group, well and dye interrogated.

Finally, you can try to get images of better quality and with the 60x magnification objective.

**Fluorescence filters (maximas):**

Hoechst 33342: Excitation: 361 nm; Emission: 497 nm.

Calcein RedOrange: Excitation: 577 nm; Emission: 590 nm

Mito Tracker Green: Excitation: 490nm; Emission: 516 nm

**2<sup>nd</sup> EXPERIMENT (Start Thursday week 1): FLUORESCENCE IMAGING OF ACTIN CYTOSKELETON**

In the 2<sup>nd</sup> experiment, the experiment is repeated, but the staining procedure is different. This time, cells are fixed, and stained for polymerized actin using rhodamine-conjugated phalloidin<sup>2</sup>.

### **Poly-D-lysine coating of glass cover slips**

Cells under investigation with fluorescence microscopy are cultured on poly-L-lysine coated cover slips. **Steps 1-4 will be performed outside the LAF bench.**

1. The cover slips (10-12 slips) are placed in a clean glass petri dish and incubated for 2 hours in 2M NaOH (aq) at room temperature.
2. The NaOH solution is removed, and the cover slips are washed twice with Milli Q water.
3. The cover slips are covered with a 100 µg/mL (H<sub>2</sub>O) poly-D-lysine solution and incubate for 10 min at room temperature.
4. The poly-L-lysine solution is removed, and 70% ethanol is added for sterilization. (For longer storages, the cover slips are placed in 98% ethanol and the glass petri dish is sealed to avoid evaporation).

### **Day 1: Seeding of DLD-1 (Team 1+2) or Caco-2 (Team 3+4) cells**

- Place a poly-L-lysine coated cover slips in 8 wells of 12 well plate. Ensure that upward facing side contains poly lysine. Let remaining alcohol evaporate.
- Use cells trypsinized in Exp 1 to seed. Seed in 8 wells of a 12-well plate, 50.000 cells per well in a total of 2 mL medium/well.

Day 2-4 is exactly the same as in Week 1.

- Use 2 wells / treatment.

### **Day 5 (Monday week 2): Fixation and fluorescent staining**

At Day 5, the cells are fixed and prepared for fluorescent staining.

**These steps will be performed outside the LAF bench.**

---

<sup>2</sup> Unless instructors feel the need to optimize parameters.

1. Inspect the cells in the microscope. Use phase contrast 10x or 20x
2. Remove all medium and quickly add 0.5mL fixative (in fume hood, 10% formalin) and incubate for 20 min at room temperature.
3. Remove the fixative and wash three times with PBS for 5 min. Discard fixative appropriately.

**From this step the petri dishes should be covered with tinfoil whenever possible.**

### **Rhodamine-Phalloidin staining**

4. Mix in an Eppendorf tube 1  $\mu$ L Rhodamine-Phalloidin to 1 mL 1% BSA in PBS, add 100  $\mu$ L per cover slip. Incubate for 30 min at room temperature. If the solution is prepared in advance, seal the Eppendorf tube with tinfoil and keep it on ice.
5. Wash the cover slips 2 x 5 min in PBS.
6. Remove cover slips from the petri dish with scalpel and mount them on microscope slides with a drop of mounting media between the slide and cell side of the cover slips.
7. Allow mounting media to cure overnight (minimum 12 hours) at room temperature. Seal the edge of the cover slip with nail polish, brand and color is optional. Let the samples dry in petri dishes covered with tinfoil and store them in the refrigerator.
8. The samples may be kept in the refrigerator for several days. If there is a need for longer storage, this may be done at -20°C.

### **Fluorescence Microscopy**

9. Microscopes:
  - a. Microscopes in room 28A.2-37 are equipped for both phase contrast and fluorescent microscopy:
    - i. Leica DM 5000 B
    - ii. Leica DM IRB
    - iii. EVOS M5000
  - b. Confocal imaging facility, 28AB.0-07, will be operated by the instructors.
10. Magnification: Use the 40x and perhaps the 60x objectives (needs oil)



11. Fluorescence filters (Leica DM IRB):
  - a. DAPI filter: Excitation: BP 340-380; Emission: LP 425,
  - b. Rhodamine filter: Excitation: BP 515-560; Emission: LP 590
12. Fading. Be aware that especially rhodamine fades quickly. Therefore, expose the areas of interest as short time as possible to fluorescent light, before acquiring images.
13. Identify the cells with phase contrast. Identify cells which are placed individually. Find the relevant focus. Switch to the fluorescence and take pictures. It is possible to take several pictures at the same position, but you might have to increase the exposure time (start with 15 ms, aperture 1).
14. Acquire the image of the DAPI staining at the exact same position (i.e. do NOT move the slide!).

## Exercise 4b – Hygiene exercise

### Experiment

This exercise illustrates some important facts of how many bacteria we have on our skin and under what circumstances we can transfer these to the surroundings.

1. Every person needs 8 LB plates w/o antibiotics.
2. Stamp each fingertip to one LB-plate and label \*Not washed\*
3. Put both index fingertips (pegefingertip) in ethanol, 70% for 30s and let air dry.  
Stamp the index fingertips onto one LB-plate and label \*Not washed, EtOH, dry\*
4. Wash hands. With wet fingers stamp another LB-plate and label \*Wash 1x, wet\*
5. Dry hands as much as possible with a paper towel and let air dry at least 1 min.  
Stamp another LB-plate and label \*Wash 1x, dry\*
6. Wash hands. With wet fingers stamp another LB-plate and label \*Wash 2x, wet\*
7. Dry hands as much as possible with a paper towel and let air dry. Stamp another LB-plate and label \*Wash 2x, dry\*
8. Wipe hands with ethanol and while hands are wet with ethanol stamp another LB-plate and label \* Wash 2x+EtOH, wet\*
9. Let hands air dry. Stamp another LB-plate and label \*Wash 2x+EtOH, dry\*
10. Incubate at 37°C O/N
11. Next day: Count and record colonies on each plate. Take pictures of each plate for documentation (use your camera phones).

## For the report - Exercise 4

- Include your laboratory notebook (scans) and annotated images, from both exercise 4a and 4b.

Immunofluorescence experiment (4a):

- Using ImageJ to quantify nucleus size based on images from the live stain and based on the fixed cells. How well do these estimated correspond to each other?
- Present the results of the nuclei and actin filament stains of DLD-1 / Caco-2 cells using text and images. Explain your results.
- What were the effects of treatment with NaBu and/or Resv on nuclei and cell size? Is it quantifiable? Discuss possible quantification strategies (even though you may not have observed obvious differences).
- What did you observe with the Mitotracker Green stain? The effect of Resv and /or NaBu?

Hygiene exercise (4b):

- What did you observe? Show the counts in a table.
- Explain the pattern of colonies.